

Operating-Boundary Routing Accountability and Transition-Window Reserve Diagnostics for Wind-Power Transitions

Junyu Li^a Juntao Du^{a,*}

^a School of Statistics and Applied Mathematics, Anhui University of Finance and Economics,
Bengbu, 233030, China

*Corresponding author.

Email addresses: ljylikezmn999@gmail.com (Junyu Li), dujuntao@aufe.edu.cn (Juntao Du)

Abstract

Wind turbine transitions from maximum power point tracking (MPPT) to blade-pitch control create concentrated reserve exposure that fleet-level forecasting accuracy metrics can miss: over-forecasts in this control window translate directly into unmet reserve requirements. We test whether supervisory control and data acquisition (SCADA) operating anchors can make a routed forecaster physically traceable enough for transition-window reserve diagnosis. The model uses a directed-diffusion gated recurrent unit with a node-level mixture-of-experts gate constrained by wind speed, blade-pitch angle, wake exposure, graph smoothness, and boundary-focused forcing. It is evaluated on the KDD Cup 2022 wind-farm SCADA benchmark (WTB), with ERA5 reanalysis retained as an observability contrast. At shortage-to-reserve cost ratio 10, gate-conditioned reserve binning reduces boundary-window violation rate from 10.38% to 9.00% and shortage energy from 3.73M to 3.19M normalized reserve-energy units relative to the same routed model with a global reserve rule, while carrying 1.85M additional reserve energy. The router recovers the MPPT-to-pitch operating partition with NMI/ARI of 0.8716 $\pm$ 0.0418 / 0.9166 $\pm$ 0.0371 across five WTB seeds, although Graph WaveNet remains the WTB accuracy reference (overall RMSE 225.74 $\pm$ 2.60). Kelmarsh/Penmanshiel external-farm runs fail the pre-specified routing criterion after local recalibration. The result is therefore a within-WTB operating-boundary diagnostic for transition-window reserve risk, together with explicit sensor, boundary-support, and held-out-routing conditions for any new-site interpretation.

Keywords: Wind-power operations; operating boundary; reserve decision; mixture of experts; regime routing; spatio-temporal graph learning.

1 Introduction

Reserve planning for wind-integrated power systems becomes most sensitive when a forecast error changes an operating decision rather than only a fleet-average error score. The maximum power point tracking (MPPT)-to-pitch transition is one such point. In the MPPT region, turbine power responds strongly to wind-speed variation; as pitch control activates near rated operation, the local power-response law changes and over-forecasts can become unmet reserve requirements. Because the transition occupies a smaller share of normal wind-farm operation, its effect can be diluted in whole-sample RMSE while still concentrating reserve exposure. A useful forecaster for this setting must therefore answer two questions together: how large is the error, and which control state generated the local response?

Spatio-temporal graph forecasters have reduced mean prediction error by modeling turbine coupling, wake interaction, and dynamic dependence [Li et al., 2018, Yu et al., 2018, Wu et al., 2019, 2020, Guo et al., 2019, Bai et al., 2020]. Wind-specific graph and sensor-fusion models add wake structure, SCADA measurements, LiDAR, and offshore-layout information to improve physical relevance [Park and Park, 2019, Yu et al., 2020, Kim et al., 2024, Daenens et al., 2025]. These models set the accuracy reference, but a low-error graph encoder does not by itself tell a reserve planner whether the current local map is MPPT-like or pitch-control-like. Mixture-of-experts (MoE) routing can separate heterogeneous response laws [Jacobs et al., 1991, Jordan and Jacobs, 1994, Shazeer et al., 2017, Fedus et al., 2022, Shi et al., 2025], yet a gate trained only through prediction loss can specialize on partitions that have no operating meaning. The missing middle layer is an auditable routing assignment that identifies which physical response law is active at the anchor time and can then be used in reserve-risk diagnosis.

We make the routing decision itself the operating-boundary diagnostic. The model encodes local spatio-temporal context with a directed-diffusion GRU and uses a node-level MoE gate to allocate fore-

casts among local experts. Wind speed, mean blade-pitch angle, wake exposure, graph smoothness, and MPPT-to-pitch boundary forcing constrain the gate so that its assignment can be compared with the declared SCADA operating partition. WTB is the source control-boundary benchmark because the aerodynamic boundary is partly hidden inside turbine-control action; ERA5 is retained as an observability contrast where the thermodynamic marker is more directly visible through sensible heat flux. The recovered gate is then connected to a validation-calibrated, test-frozen reserve diagnostic that reports cost, violation rate, reserve energy, and shortage energy under declared shortage-to-reserve cost ratios.

This paper makes three contributions. First, gate-conditioned reserve binning turns the recovered operating assignment into a transition-window reserve diagnostic: at cost ratio 10 it reduces same-model boundary-window violation from 10.38% to 9.00% and shortage energy from 3.73M to 3.19M normalized units, while adding 1.85M reserve energy. Second, SCADA-anchored routing recovers the WTB MPPT-to-pitch partition with NMI/ARI of 0.8716 $\pm$ 0.0418 / 0.9166 $\pm$ 0.0371 across five seeds, and the agreement collapses under boundary-anchor removal while surviving placebo, spatial-holdout, and future-period stress checks. Third, the Kelmarsh/Penmanshiel external audit is reported as a deployment-safety result: it fails the pre-specified held-out routing criterion and identifies the sensor coverage, pitch observability, turbine geometry, and boundary-support conditions that must be locally verified before a new wind farm can use a gate-bin reserve rule.

2 Related Work

2.1 From average wind-power accuracy to transition-window risk

Wind-power forecasting has moved from single-site point prediction toward models that encode the physical sources of error. Ramp and process reviews identify non-stationary weather and rapid power changes as recurrent operational risks [Pinson, 2013, Gallego-Castillo et al., 2015, Yang et al., 2025], while uncertainty reviews clarify why point accuracy alone is insufficient for decisions that consume forecasts [Wang et al., 2025, Haq et al., 2025]. Spatio-temporal graph models then supply the main accuracy reference: diffusion and graph convolution encode directed or local dependence [Li et al., 2018, Yu et al., 2018, Wu et al., 2019], and adaptive or attention-based graph encoders relax fixed-neighborhood assumptions [Guo et al., 2019, Bai et al., 2020, Wu et al., 2020]. Wind-specific graph studies bring the same machinery back to turbines by adding wake structure, SCADA fields, LiDAR, and offshore layouts [Park and Park, 2019, Yu et al., 2020, Kim et al., 2024, Daenens et al., 2025]. Together, this literature defines the low-error benchmark. Its remaining blind spot is operational: average accuracy does not identify the control-transition samples where reserve exposure is concentrated. Addressing that blind spot requires the routing assignment itself to carry operating-state meaning.

2.2 Operating regimes, SCADA observability, and physical interpretability

Regime-aware modeling separates local response laws, but that separation is useful for operations only when the gate can be interpreted. Classical adaptive and hierarchical MoE models introduced input-dependent expert assignment [Jacobs et al., 1991, Jordan and Jacobs, 1994]. Sparse modern MoE systems made routing scalable, while also showing that collapse, starvation, and load-allocation instability must be controlled during training [Shazeer et al., 2017, Fedus et al., 2022]. Recent time-series routers extend specialization to sequence forecasting [Shi et al., 2025, Cao and Wang, 2026, Tian et al., 2026], but prediction loss by itself need not produce a physically meaningful partition. Physics-guided learning offers the missing constraint principle: domain knowledge can enter features, architectures, losses, probabilities, or diagnostic checks [Karpatne et al., 2017, Read et al., 2019, Raissi et al., 2019, Karniadakis et al., 2021, Zehtabiyani-Rezaie et al., 2023, Parsa, 2025, Gao et al., 2025]. For wind turbines, that principle has to pass through SCADA observability: operating-state labels depend on wind speed, pitch channels, active power, status masks, and turbine-specific power curves [Tautz-Weinert and Watson, 2017, Zhou et al., 2024]. This paper therefore constrains the routing decision itself—the assignment of an operating

instant to a local predictor—so downstream reserve logic can consume a physically traceable gate.

2.3 Forecast-driven reserve allocation and risk diagnostics

Forecast value in power systems is realized through reserve, commitment, balancing, and trading decisions. Reserve studies establish the operational premise: variable generation changes operating-reserve requirements, and reserve demand should respond to uncertainty rather than follow a fixed margin [Doherty and O'Malley, 2005, Ela et al., 2011]. Quantile and probabilistic wind-forecasting methods provide the statistical bridge from point forecasts to decision risk [Bremnes, 2004, Nielsen et al., 2006, Zhang et al., 2014, Wang et al., 2025]. Market and unit-commitment studies then show how forecast uncertainty becomes imbalance exposure, reserve cost, and commitment risk [Pinson et al., 2007, Wang et al., 2011, Zhou et al., 2013]. This literature usually asks how much reserve should be carried given a forecast or uncertainty estimate. The present work asks the earlier diagnostic question: can the model identify the operating window in which reserve exposure is concentrated? That framing makes the reserve audit complementary to probabilistic dispatch and market-clearing studies rather than a substitute for them.

3 Methodology

3.1 Problem setup and notation

The forecasting task is written to separate two decisions that dense models often merge: predicting the future and deciding which local mapping should be active at the anchor time. Let $G = (V, E)$ denote a spatial graph with $N = |V|$ nodes. For each node $i \in V$ and time step t , we observe a feature vector $\mathbf{x}_{i,t} \in \mathbb{R}^F$ and predict a scalar target $y_{i,t} \in \mathbb{R}$. Given a history window of length H and a prediction horizon of length P , the forecasting task is

$$\hat{\mathbf{Y}}_{t+1:t+P} = \mathcal{F}(\mathbf{X}_{t-H+1:t}, \mathcal{A}_{t-H+1:t}),$$

where $\mathbf{X}_{t-H+1:t} \in \mathbb{R}^{H \times N \times F}$ and $\mathcal{A}_{t-H+1:t}$ denotes either a time-varying directed graph sequence (WTB) or a static graph repeated over time (ERA5). Invalid or missing targets are excluded by a supervision mask.

The information boundary is strict. All inputs, graph weights, regime anchors, and gate anchors are observed no later than the forecast issue time t ; supervised targets begin at $t + 1$. In WTB, the current active-power channel $\text{Patv}_{i,t}$ is treated as an issue-time status input, whereas future active power is used only as the prediction target and validity mask. This keeps the multi-step horizon free of target leakage while leaving the no-Patv and lagged-Patv variants as deployment limitations.

Routing is node-level: each node at each anchor time receives its own gate distribution $\mathbf{g}_{i,t}$. This matters because nearby turbines or grid cells can occupy different local regimes within the same sequence. A notation table is provided in Appendix A.

3.2 Operating-boundary anchors and labels

The gate is not supervised everywhere. It is anchored where the physical interpretation is clearest, while ambiguous samples remain governed by prediction loss and routing regularization. This engineering layer defines what counts as a control regime and when that regime is observable from the data.

3.2.1 WTB operating regimes

In WTB, the primary operating boundary is the transition from MPPT to pitch control. Let

$$\bar{p}_{i,t} = \frac{1}{3} (p_{i,t}^{(1)} + p_{i,t}^{(2)} + p_{i,t}^{(3)}).$$

The WTB column name Pab denotes blade pitch angle. We use $\bar{p}_{i,t}$, also reported as Pab_mean in tables and figures, for the three-blade pitch average. The regime labels use wind speed and mean pitch angle at the anchor time, not future active power.

Using a cut-in threshold u_{idle} , a pitch-transition threshold u_{rated} , and a pitch-angle threshold p_{th} , we define

$$R_{i,t}^{\text{wtb}} = \begin{cases} 0, & \text{if } \text{Wspd}_{i,t} < u_{\text{idle}} \quad (\text{idle}), \\ 1, & \text{if } u_{\text{idle}} \leq \text{Wspd}_{i,t} \leq u_{\text{rated}} \wedge \bar{p}_{i,t} < p_{\text{th}} \quad (\text{MPPT}), \\ 2, & \text{if } \text{Wspd}_{i,t} > u_{\text{rated}} \wedge \bar{p}_{i,t} \geq p_{\text{th}} \quad (\text{pitch-control}), \\ 3, & \text{otherwise} \quad (\text{transition}). \end{cases}$$

Only the first three classes are used in direct alignment. Transition samples are retained for analysis but masked out of label supervision through

$$M_{i,t}^{\text{wtb}} = \mathbf{1}[R_{i,t}^{\text{wtb}} \neq 3].$$

The reported implementation uses $u_{\text{idle}} = 3.0 \text{ m s}^{-1}$, $u_{\text{rated}} = 10.5 \text{ m s}^{-1}$, and $p_{\text{th}} = 2.0^\circ$ (Appendix A). These thresholds are operating anchors rather than universal turbine constants.

3.2.2 ERA5 thermodynamic regimes

In ERA5, the anchor is built around the stable-to-convective transition. Surface sensible heat flux changes sign across that transition, and large flux gradients mark disturbed periods. Let $\Delta \text{sshf}_{i,t} = \text{sshf}_{i,t} - \text{sshf}_{i,t-1}$. We use a small-margin threshold $\varepsilon_{\text{sshf}}$ for $|\text{sshf}|$. We also use a high-quantile threshold $q_{0.95}^\Delta$ for $|\Delta \text{sshf}|$. The regime label is then defined by

$$R_{i,t}^{\text{era5}} = \begin{cases} 0, & \text{if } \text{sshf}_{i,t} > \varepsilon_{\text{sshf}} \wedge |\Delta \text{sshf}_{i,t}| \leq q_{0.95}^\Delta \quad (\text{convective}), \\ 1, & \text{if } \text{sshf}_{i,t} < -\varepsilon_{\text{sshf}} \wedge |\Delta \text{sshf}_{i,t}| \leq q_{0.95}^\Delta \quad (\text{stable}), \\ 2, & \text{otherwise} \quad (\text{transition/anomalous}). \end{cases}$$

The exact thresholds used in the reported experiments are listed in Appendix A.

3.3 Shared architecture

3.3.1 Physical graph construction

The graph is part of the physical problem specification. WTB uses a directed wake graph: candidate turbine pairs are filtered by proximity, activated when an upstream turbine lies inside a wind-aligned downstream cone, weighted by streamwise and cross-stream decay, and pruned to the strongest inbound neighbors. The same directed weights define a wake score, which supplies the WTB wake auxiliary label on MPPT and pitch-control samples. ERA5 uses a symmetric Haversine-Gaussian k_{nn} graph because the thermodynamic regime marker is already visible in the observed state. Appendix A gives the graph equations and constants.

3.3.2 Directed-diffusion GRU encoder and node-level gate

The encoder is kept modest so that changes in performance can be traced to routing and regularization. Each input window is concatenated with its feature-missingness mask before projection. Two directed diffusion blocks then aggregate self, inbound, and outbound messages on each graph snapshot. Let $\mathbf{X}_t^{(\ell)}$ denote the node state at layer ℓ and time t . A diffusion block updates each node by

$$\mathbf{X}_t^{(\ell+1)} = \text{LN}\left(\mathbf{W}_{\text{self}}^{(\ell)}\mathbf{X}_t^{(\ell)} + \mathbf{W}_{\text{in}}^{(\ell)}\text{Agg}_{\text{in}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{W}_{\text{out}}^{(\ell)}\text{Agg}_{\text{out}}(\mathbf{X}_t^{(\ell)}, \mathcal{A}_t) + \mathbf{R}^{(\ell)}\mathbf{X}_t^{(\ell)}\right),$$

where Agg_{in} and Agg_{out} are weighted sums over inbound and outbound neighbors. The spatially updated sequence is then processed by a GRU to produce a node-wise context vector $\mathbf{h}_{i,t}$.

Each expert head f_e maps $\mathbf{h}_{i,t}$ to a P -step forecast. Routing remains dense during the forward pass. Every expert contributes, but its contribution is weighted by the soft gate distribution. The gate receives both the learned context and a small physics anchor,

$$\mathbf{z}_{i,t} = \text{MLP}_{\text{gate}}([\mathbf{h}_{i,t}; \mathbf{a}_{i,t}]), \quad \mathbf{g}_{i,t} = \text{softmax}\left(\frac{\mathbf{z}_{i,t}}{\tau}\right),$$

and the forecast is

$$\hat{\mathbf{y}}_{i,t+1:t+P} = \sum_{e=1}^E g_{i,t}^{(e)} f_e(\mathbf{h}_{i,t}).$$

WTB uses the anchor

$$\mathbf{a}_{i,t}^{\text{WTB}} = [\text{Wspd}_{i,t}, \text{Pab_mean}_{i,t}, s_{i,t}^{\text{wake}}, \text{Patv}_{i,t}],$$

whereas ERA5 uses

$$\mathbf{a}_{i,t}^{\text{ERA5}} = [\text{sshf}_{i,t}, \text{t2m}_{i,t}, \text{wind_speed}_{i,t}, \Delta \text{sshf}_{i,t}].$$

The difference reflects the observability contrast: WTB needs help to recover a partly hidden operating boundary, whereas ERA5 already exposes the relevant thermodynamic marker in the state. The active-power anchor is standardized from training-split statistics and is never filled from the prediction horizon.

3.3.3 Routing stack and evidence boundary

This makes the routing stack an anchor-constrained diagnostic rather than an unsupervised operating-state discovery method. High gate-regime agreement means that the implemented router obeys the declared SCADA boundary under the available anchor set; it does not establish anchor-free regime recovery.

3.4 Physics-aligned routing regularizers

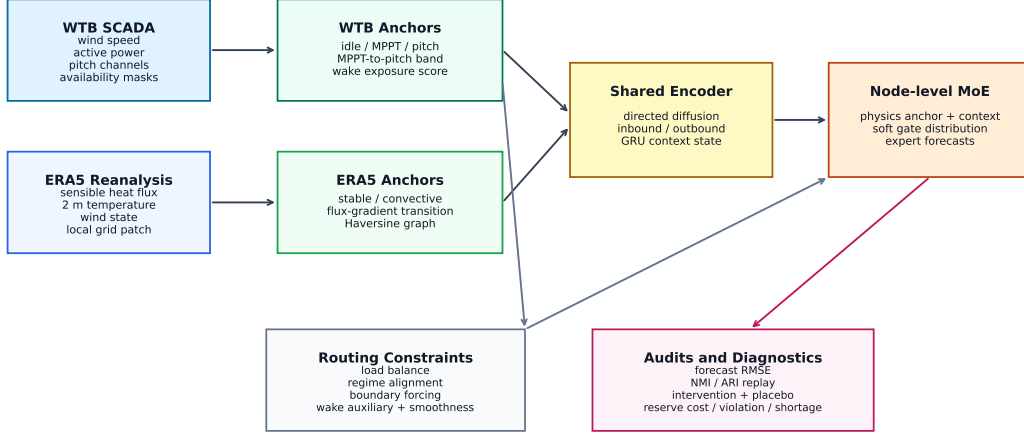
Prediction loss does not uniquely identify the gate when several experts can fit averaged dynamics similarly well [Shazeer et al., 2017, Fedus et al., 2022]. WTB makes this failure visible: inflow variation, wake disturbance, and pitch control are observed together, so a low-loss router can still ignore the operating boundary. We use routing regularizers for the specific failure modes that matter here. The total objective is

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{bal}} \mathcal{L}_{\text{bal}} + \lambda_{\text{align}} \mathcal{L}_{\text{align}} + \lambda_{\text{force}} \mathcal{L}_{\text{force}} + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}} + \lambda_{\text{smooth}} \mathcal{L}_{\text{smooth}},$$

with inactive terms set to zero in settings where they are not used. Exact weights are listed in the Appendix.

Each mini-batch encodes the history window and graph sequence, forms node-level routing probabilities from the learned context and physics anchor, and aggregates expert forecasts. The prediction loss is always active. Routing penalties are then added only in settings where they are declared. Checkpoint selection remains based on validation RMSE, so the penalties constrain responsibility without becoming the validation-selection metric.

Physics-aligned regime-aware MoE for operating-boundary accountability



Interpretation: WTB tests a control-confounded turbine boundary; ERA5 is an observability contrast where the regime marker is visible.
The routed model is evaluated as an accountability object, then linked to a frozen validation-to-test reserve diagnostic.

Figure 1: Physics-aligned regime-aware MoE. The two datasets share the same directed-diffusion GRU encoder and node-level MoE routing mechanism. WTB uses a dynamic wake graph and a boundary-focused forcing term, whereas ERA5 uses a Haversine-Gaussian graph and a thermodynamic regime anchor.

3.4.1 Prediction objective

The prediction loss remains the anchor of the optimization, because a routing correction that stops serving the forecast defeats the purpose of the model. We retain the masked mean-squared-error loss

$$\mathcal{L}_{\text{pred}} = \frac{\sum_{i,t,\tau} m_{i,t+\tau}^y (\hat{y}_{i,t+\tau} - y_{i,t+\tau})^2}{\sum_{i,t,\tau} m_{i,t+\tau}^y},$$

where $m_{i,t+\tau}^y$ is the target-validity mask. All routing regularizers remain subordinate to this prediction objective.

3.4.2 Load balancing

A balancing term prevents early expert collapse before regime-specific structure has time to emerge. It couples hard top- K usage with soft probability mass over valid node-time samples, following the practical role of Switch-style load penalties but applied to node-level routing. The term stabilizes training; physical ownership is supplied only by the anchored alignment terms below. Appendix A gives the exact expression.

3.4.3 Regime-anchor alignment

Balanced expert usage can still produce a physically meaningless partition. The alignment term connects the first C gate logits to the coarsest regime structure that can be identified with high confidence: idle, MPPT, and pitch-control for WTB; stable, transition, and convective for ERA5. Let $\mathbf{z}_{i,t}^{(1:C)}$ denote these logits, and let $R_{i,t}$ and $M_{i,t}$ be the primary regime label and validity mask. The alignment loss is

$$\mathcal{L}_{\text{align}} = \frac{1}{|\Omega|} \sum_{(i,t) \in \Omega} \text{CE}(\mathbf{z}_{i,t}^{(1:C)}, R_{i,t}), \quad \Omega = \{(i,t) : M_{i,t} = 1\},$$

with inverse-frequency class weights estimated on the training split. The fixed index convention breaks MoE label-permutation symmetry for the anchored logits during training, so cross-seed statements such as “MPPT-aligned” or “pitch-aligned” refer to this declared mapping rather than to post-hoc expert relabeling.

3.4.4 Boundary-focused forcing (WTB only)

The MPPT-to-pitch boundary in WTB remains ambiguous even after coarse regime alignment because control action partly masks the mechanical transition. A focused forcing term acts only on the most informative MPPT and pitch-control samples. Let

$$Y_{i,t}^{\text{force}} = \begin{cases} 0, & R_{i,t}^{\text{wtb}} = 1, \\ 1, & R_{i,t}^{\text{wtb}} = 2. \end{cases}$$

Using the MPPT-aligned and pitch-aligned expert logits fixed above, we define

$$\mathcal{L}_{\text{force}} = \frac{1}{|\Omega_{\text{force}}|} \sum_{(i,t) \in \Omega_{\text{force}}} \text{CE} \left(\begin{bmatrix} z_{i,t}^{(\text{mppt})} \\ z_{i,t}^{(\text{pitch})} \end{bmatrix}, Y_{i,t}^{\text{force}} \right),$$

where $\Omega_{\text{force}} = \{(i, t) : R_{i,t}^{\text{wtb}} \in \{1, 2\}, M_{i,t} = 1\}$. This term concentrates the intervention on high-confidence MPPT and pitch-control anchors already visible at the issue time. It increases routing identifiability around the boundary after the operating state is expressed in the SCADA channels; it is not designed or evaluated as a lead-time switch predictor.

3.4.5 Wake auxiliary supervision and graph smoothness

After the MPPT-to-pitch boundary is repaired, wake-sensitive samples can still be mixed with non-wake samples inside the same operating regime. A wake auxiliary label identifies that residual ambiguity in WTB, while graph smoothness discourages noisy neighbor-to-neighbor gate jumps in both settings. The auxiliary term is active only for MPPT and pitch-control samples with a defined wake flag; the smoothness term is computed on the graph snapshot associated with the anchor time. Appendix A reports both formulas.

3.5 Operational reserve diagnostic protocol

The reserve diagnostic tests whether a physically auditable gate changes the reserve tradeoff in the MPPT-to-pitch window after the point-forecast model has been trained. For each saved WTB run, validation predictions select reserve levels; test predictions are then evaluated once with those levels fixed. The protocol follows the same information boundary as an operator who calibrates a short-term reserve rule before using it on a later period.

Let $\hat{y}_{i,t}$ be the scheduled point forecast and $y_{i,t}$ the realized active power at an evaluated horizon cell. We define over-forecast shortfall as

$$s_{i,t} = \max(\hat{y}_{i,t} - y_{i,t}, 0),$$

because this is the error direction that leaves a reserve schedule exposed. For a reserve policy b and empirical quantile q , the validation split estimates a reserve level

$$r_{b,q} = Q_q(\{s_{i,t} : (i, t) \in b, (i, t) \in \mathcal{V}\}),$$

where Q_q is an empirical quantile and $\mathcal{V}$ is the validation split. The grid is fixed before test evaluation:

$$q \in \{0.50, 0.60, 0.70, 0.80, 0.85, 0.90, 0.95, 0.975, 0.99\}.$$

The reserve bin b is either the whole sample, a physical operating bin, or a gate-derived bin. For each shortage-to-reserve cost ratio ρ , the protocol selects the validation quantile that minimizes

$$C_{\mathcal{T}}(b, q; \rho) = \sum_{(i,t) \in b \cap \mathcal{T}} [r_{b,q} + \rho \max(s_{i,t} - r_{b,q}, 0)] \Delta t.$$

The selected (b, q) rule is then frozen and applied to the test split. The reported test metrics are total cost C , violation rate $\Pr[s_{i,t} > r_b]$, reserve energy $\sum r_b \Delta t$, and shortage energy $\sum \max(s_{i,t} - r_b, 0) \Delta t$. All totals are reported in normalized reserve-energy cost units. Thus a value written as 84.58M denotes 84.58 million normalized reserve-energy cost units, not currency.

The cost ratio ρ is an energy-system assumption rather than an abstract tuning knob. If the marginal cost of carrying one unit of reserve energy is C_r , then $\rho = 10$ charges one unit of residual shortage as $10C_r$. Ratios 5–10 represent moderate reliability settings such as transition-window scheduling or imbalance screening; ratios 20–50 represent scarcity-aware screening where shortage avoidance dominates local cost savings. Four policies are evaluated: Graph WaveNet/global, Graph WaveNet/physical-bin, boundary router/global, and boundary router/gate-bin. Only same-model global comparisons attribute gate-bin reserve effects to the learned router.

Reserve diagnostic algorithm. Input saved validation/test point forecasts, targets, masks, physical anchors, and gate outputs. Compute over-forecast shortfall on valid cells. Form global, physical-bin, and gate-bin reserve groups on validation outputs only. For each cost ratio and group, select the empirical shortfall quantile that minimizes validation reserve-plus-shortage cost. Freeze the selected reserve levels and evaluate them once on test cells. Aggregate cost, violation rate, reserve energy, and shortage energy over full, boundary, transition, ramp, horizon-bucket, and time-of-day slices. Report horizon-specific empirical-quantile rows only as a post-hoc what-if; no probabilistic model is trained.

4 Experimental Setup

4.1 Datasets and preprocessing

The experiment uses two observability settings, not two interchangeable benchmarks. WTB is the control-confounded case. The KDD Cup 2022 benchmark contains 134 turbines and 245 days of 10 min SCADA measurements [Zhou et al., 2024]. The MPPT-to-pitch transition is only indirectly visible because inflow variation, wake disturbance, and blade-pitch control are mixed in the same stream.

The data are reorganized into synchronous tensors with $T = 35,280$ time steps, $N = 134$ turbines, and $F = 11$ dynamic inputs. Missing inputs are forward-filled within turbine and then completed with training-set means. Invalid or non-positive active power is retained only through its mask so that the model still sees operational irregularity without treating those points as supervised targets.

ERA5 is the signal-expressive case. We use three archived months over a fixed 16×16 hourly patch [Hersbach et al., 2020]. Sensible heat flux and its temporal variation make the stable-to-convective transition directly observable. The retained tensor has $T = 2208$ frames, $N = 256$ nodes, and eight input variables. Both datasets use the same history and horizon lengths, $H = 36$ and $P = 24$, so differences in performance cannot be attributed to unequal context length. The chronological split is 180/30/35 days for WTB and 1325/441/442 frames for ERA5.

4.2 In-family comparison protocol

The comparison has two layers. The first isolates the routing mechanism after shared-capacity explanations have been controlled. The dense baseline uses the same directed-diffusion GRU encoder and replaces the expert mixture with one dense prediction head. The unconstrained MoE adds routed capacity without physical correction. The corrected-routing comparator is dataset-specific: ERA5 uses the

Table 2. In-family model comparison used for mechanism validation in the current encoder family.

Model	Routing	Extra terms	Params	Role
Dense (matched)	Single dense head	None	WTB 110,012 / ERA5 103,272	Shared-mapping reference
Unconstrained MoE	Node-level soft gate	Prediction loss only	WTB 110,012 / ERA5 103,843	Tests whether routed capacity alone is enough
Corrected routing comparator	Node-level soft gate	WTB boundary-forced: $L_{bal} + L_{align} + L_{force}$; ERA5: full corrected stack	WTB 110,012 / ERA5 103,843	Dataset-specific corrected-routing comparator

full corrected stack, while WTB uses the boundary-focused variant selected for gate recovery. Parameter budgets are matched in WTB and near-matched in ERA5, where the dense model has 103,272 parameters and the routed models have 103,843 parameters.

The second layer tests the RMSE price against stronger and more engineering-facing baselines. Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST are included for WTB, together with deterministic persistence, a fitted physical power-curve baseline, XGBoost and LightGBM lag-feature predictors, and a DLinear-style LTSF baseline. Graph WaveNet, STGCN, PatchTST, TCN, and a deterministic persistence predictor are included for ERA5. This split avoids using one comparison for two different claims. The in-family layer tests whether physical routing changes the learned partition under a fixed backbone. The baseline layer tests whether the routing intervention remains honest about forecast error when compared with established temporal, graph, and engineering predictors.

For compactness, the mechanism tables shorten the three in-family model names to **Dense (matched)**, **Unconstrained MoE**, and **Corrected routing comparator**. The WTB forecasting tables additionally report Graph WaveNet, Graph Transformer, GAT-GRU, PatchTST, the full physics-aligned MoE, the WTB boundary-forced router, and engineering baselines built from persistence, power curves, gradient-boosted lag features, and a DLinear-style LTSF readout. The ERA5 forecasting tables additionally report persistence and four learned external baselines. The in-family mechanism families are summarized in Table 2.

4.3 Training protocol and evaluation metrics

All neural models use the same training protocol where the architecture permits it. We use AdamW, early stopping on validation RMSE, gradient clipping, and mixed-precision training on a single CUDA-enabled GPU. Exact optimizer constants are reported in the Appendix so that the main text can stay focused on the comparison logic. The WTB dense baseline, Unconstrained MoE, full physics-aligned MoE, and boundary-forced router are repeated across five strict-mask seeds. The synchronized strong-baseline refresh covers Graph WaveNet, Graph Transformer, GAT-GRU, and PatchTST across seeds 201–205, giving 20 completed baseline runs under the same strict anchor-valid evaluation cache. The engineering WTB baselines are deterministic or sampled single-run readouts from the same frozen cache and are used to anchor practical forecast difficulty rather than to make a seed-level neural ranking. ERA5 main in-family models are repeated across five seeds, and the ERA5 learned baselines are repeated across three seeds. Thresholds and routing weights are set before test evaluation and then audited by sensitivity sweeps and boundary negative controls, so the main routing result is not selected by test-set NMI.

The metrics follow the claim. Overall MAE and RMSE measure forecasting accuracy. Switch-window MAE and RMSE use the evaluator’s transition-window mask around regime changes. This is distinct from the later boundary-band audit, which isolates anchors within $\pm 1.0 \text{ m s}^{-1}$ of the WTB rated-wind

MPPT-to-pitch boundary. Regime-wise RMSE locates the error reduction. Normalized mutual information (NMI), adjusted Rand index (ARI), usage entropy, and confusion matrices test whether the gate corresponds to physically interpretable structure.

Figure 2 gives the operating-decision context before the forecasting results are introduced: graph geometry shows where forecast errors propagate, and the regime-anchor panels show which sensor-derived boundaries can support reserve diagnostics.

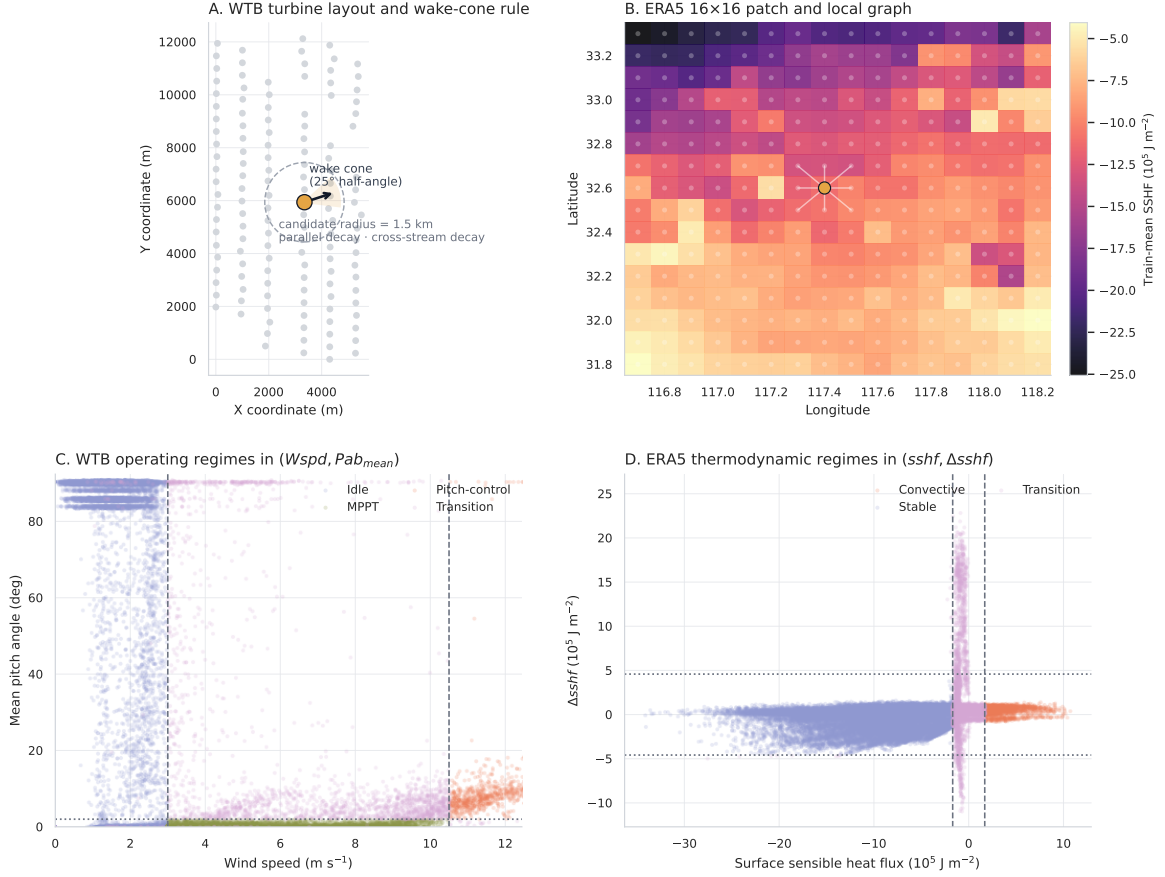


Figure 2: Operating-decision context and physical regime anchors. (A) WTB turbine layout with the schematic wake cone and retained candidate radius used in the dynamic directed wake graph. (B) ERA5 16x16 patch with training-mean sensible heat flux and local Haversine-Gaussian graph connections around the central node. (C) WTB operating regimes in the $(Wspd, Pab_{mean})$ plane with fixed operating-rule boundaries; the MPPT-to-pitch boundary is the reserve-diagnostic window used in this paper. (D) ERA5 thermodynamic regimes in the $(sshf, \Delta sshf)$ plane with thresholds estimated from the training split, included as an observability contrast.

5 Results and Discussion

5.1 Forecasting Accuracy: Establishing the Baseline

Strong graph and lag-feature baselines establish an accuracy ceiling that the physics-constrained router does not match. Graph WaveNet reaches overall RMSE 225.74 $\pm$ 2.60 on WTB, and LightGBM/XGBoost lag-feature baselines are similarly competitive at overall RMSE 227.12 and 227.88. The boundary-forced router reaches overall RMSE 236.13 $\pm$ 8.41, a 10.39-unit penalty relative to Graph WaveNet. This gap quantifies the main engineering tradeoff: the router is not selected for whole-sample error minimization, but for accountable state assignment at an operating boundary.

Table. WTB engineering forecasting baselines evaluated on the strict cache.

Baseline	Overall RMSE	Switch RMSE	Pitch-control RMSE
Persistence	243.23	246.61	262.35
Physical power curve	241.02	245.37	299.30
XGBoost lag-feature	227.88	233.07	277.33
LightGBM lag-feature	227.12	232.24	270.23
DLinear-style LTSF	246.81	251.15	284.06

ERA5 reinforces the same distinction. Persistence is the strongest headline reference, while Graph WaveNet is the lowest-RMSE learned model. Physics-aligned routing improves regime agreement relative to an unconstrained MoE, but it does not turn the learned model into the lowest-error ERA5 predictor. The forecasting results therefore set the terms for the rest of the article: any value of routing correction must appear through operating-regime accountability and reserve-risk diagnosis, not through a general forecasting leaderboard win.

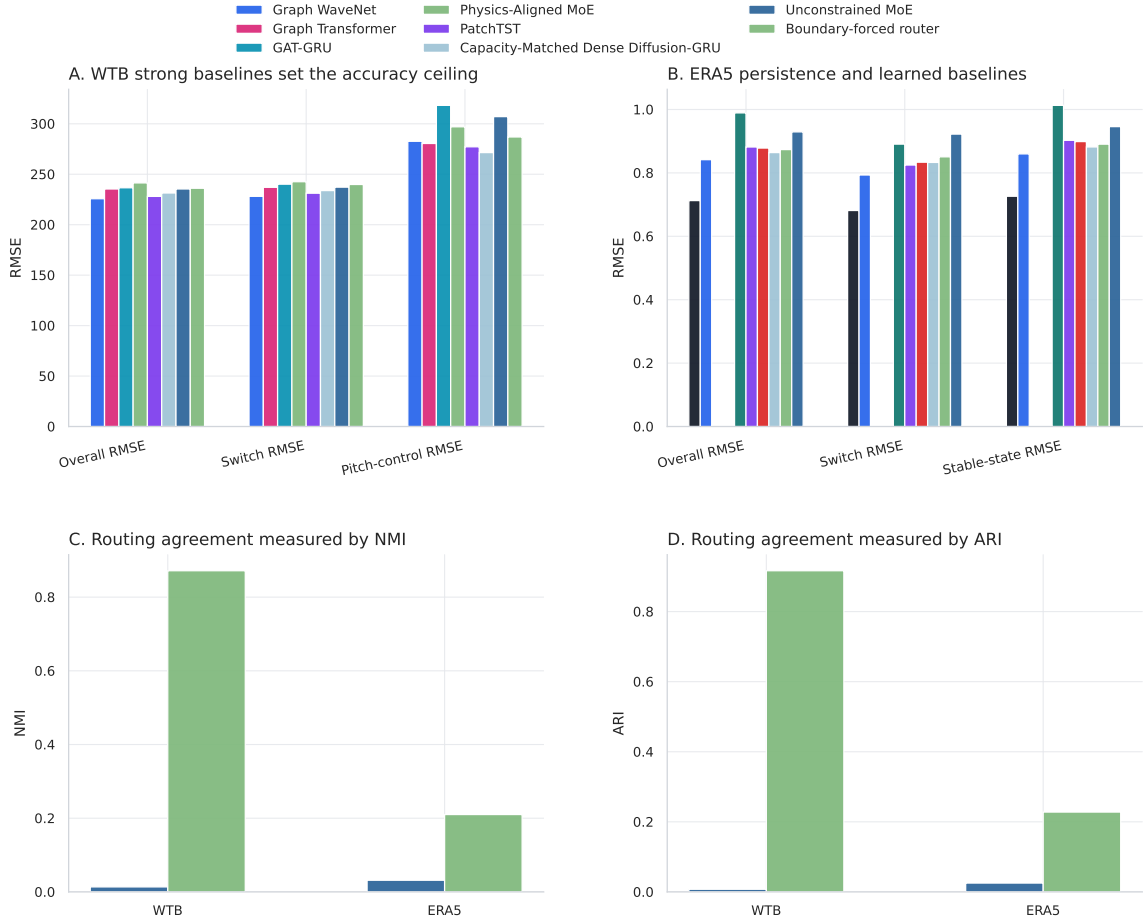


Figure 3: Summary of forecasting and routing outcomes. Panels A and B compress the learned-model error comparison for WTB and ERA5, with ERA5 persistence shown as a deterministic reference. In WTB, the displayed corrected comparator is the boundary-forced routing variant; the deeper full stack is reported separately. Panels C and D summarize routing agreement through NMI and ARI.

5.2 Transition-Window and Regime-Slice Behavior

The MPPT-to-pitch boundary is the hardest WTB forecasting slice, which is exactly why average RMSE is not enough. In the strict boundary-band audit, anchors within $\pm 1.0 \text{ m s}^{-1}$ of the rated-wind boundary

Table 1. Strict-mask WTB MPPT-to-pitch boundary-slice audit.

Slice	RMSE	NMI	ARI	Anchor cells
boundary_band	321.1711 +/- 21.3067	0.6562	0.7766	7879
nonboundary_valid	254.0981 +/- 8.6588	0.6326	0.6056	283797
mppt_core	253.7099 +/- 8.7181	NA	NA	277719
pitch_core	271.1653 +/- 16.9980	NA	NA	6175

Table 2. Strict-anchor-mask WTB five-seed metrics.

Seed	Overall RMSE	Switch RMSE	Pitch RMSE	NMI	ARI
201	245.0565	246.9875	308.0279	0.8685	0.9242
202	244.3905	248.7028	281.8781	0.9377	0.9692
203	228.0898	229.0400	269.9951	0.8221	0.8658
204	227.7822	232.0819	263.0312	0.8591	0.9060
205	235.3335	242.4639	311.8080	0.8708	0.9177
Mean +/- SD	236.1305 +/- 8.4096	239.8552 +/- 8.8508	286.9480 +/- 22.0652	0.8716 +/- 0.0418	0.9166 +/- 0.0371

have RMSE 321.17 +/- 21.31, which is 67.07 higher than valid non-boundary MPPT/pitch anchors and roughly 50–67 units higher than the single-regime MPPT and pitch-control cores. The same boundary band still has measurable mixed-regime structure, with NMI/ARI 0.6562 / 0.7766. The transition window is therefore both a forecasting stress point and a meaningful routing target.

The broader regime slices keep the accuracy-accountability tradeoff visible. Graph WaveNet and PatchTST remain strong references in switch-window and pitch-control RMSE, while the boundary-forced router pays an error cost. That cost is not a defect hidden by the analysis; it is the price of forcing the model to keep responsibility aligned with a declared physical partition. The question becomes whether that partition changes the reserve tradeoff where the control law changes.

5.3 Physics-Aligned Gate Recovery

Physics-anchored routing substantially recovers the WTB MPPT-to-pitch operating partition. Across five strict-mask WTB seeds, the boundary-forced router reaches gate-regime NMI/ARI of 0.8716 +/- 0.0418 / 0.9166 +/- 0.0371. ERA5 shows a weaker but directionally consistent pattern: physics-aligned routing raises NMI/ARI from 0.0317 +/- 0.0219 / 0.0253 +/- 0.0626 for Unconstrained MoE to 0.2100 +/- 0.0972 / 0.2279 +/- 0.1296. This contrast is informative. When the regime marker is visible in the input state, routing correction calibrates an already expressed partition; when turbine-control action partly hides the boundary, stronger SCADA anchoring is needed.

The deterministic anchor-only router is a strong semantic baseline, not a straw comparison. It reaches slightly higher NMI/ARI than the trainable boundary-forced router (0.8895/0.9219 versus 0.8716/0.9166) and lower headline RMSE. The trainable router is justified only under a narrower requirement: the responsibility assignment must remain part of the forecast and reserve evaluation. Under 0.5-standard-deviation boundary-anchor noise, anchor-only overall RMSE degrades by 105.56, whereas the trainable router degrades by 5.78. Thus the learned gate is not a better fixed labeler; it is a more robust responsibility layer for an operating diagnostic.

Figure 4 makes the WTB repair visible in the operating plane. After correction, dominant expert regions align with the MPPT and pitch-control structure, and the confusion matrices contract around the intended partition. Figure 5 adds the temporal view: the WTB gate carries responsibility through a switch window, while the ERA5 gate evolves with sensible heat flux and regime shading. Together, these panels explain why the same routing prior has different value across the two observability settings.

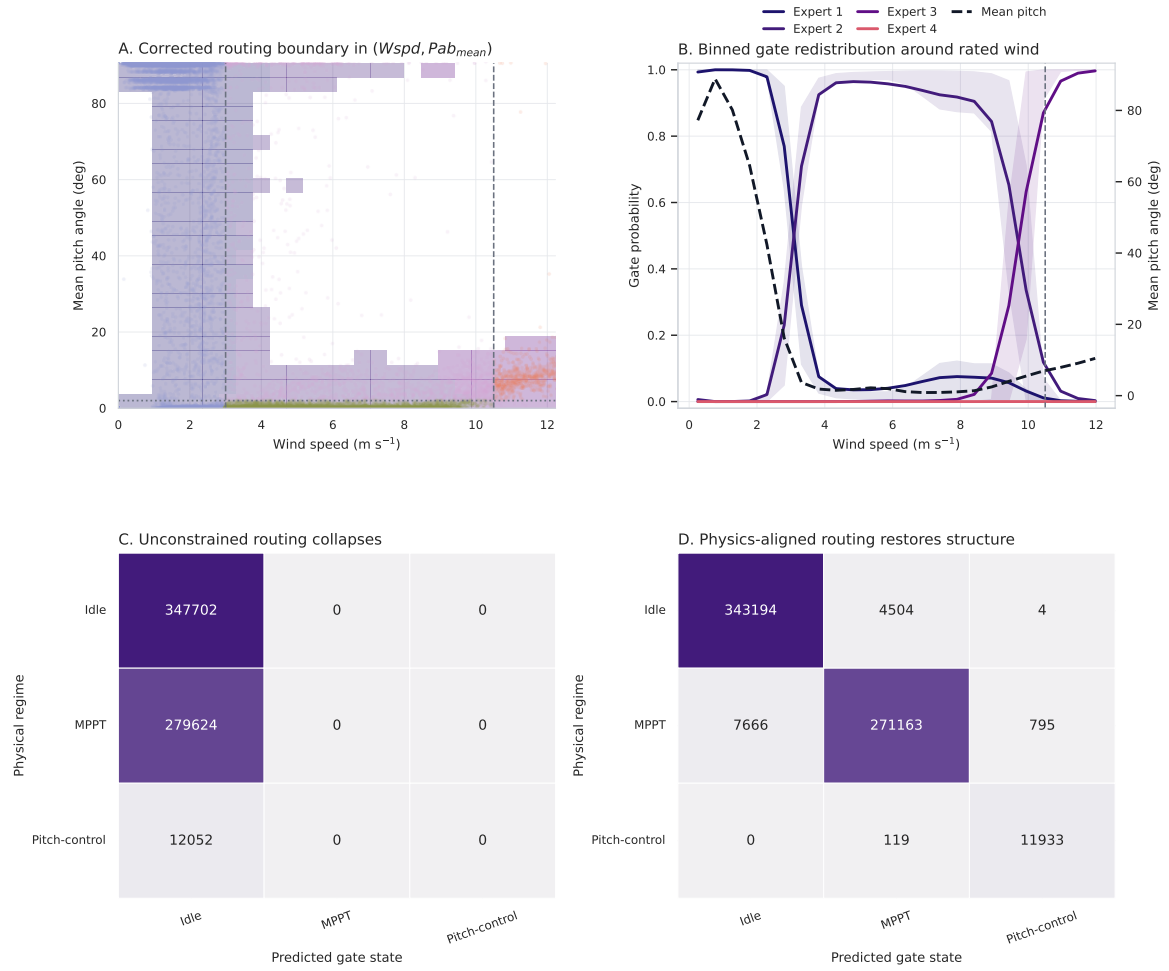


Figure 4: WTB routing diagnostics. Panel A maps the dominant expert in the $(Wspd, Pab_{mean})$ plane for the boundary-forced routing variant, with the sample cloud colored by physical operating regime. Panel B shows the binned gate redistribution against wind speed together with the mean pitch curve. Panels C and D compare the unconstrained and corrected confusion matrices.

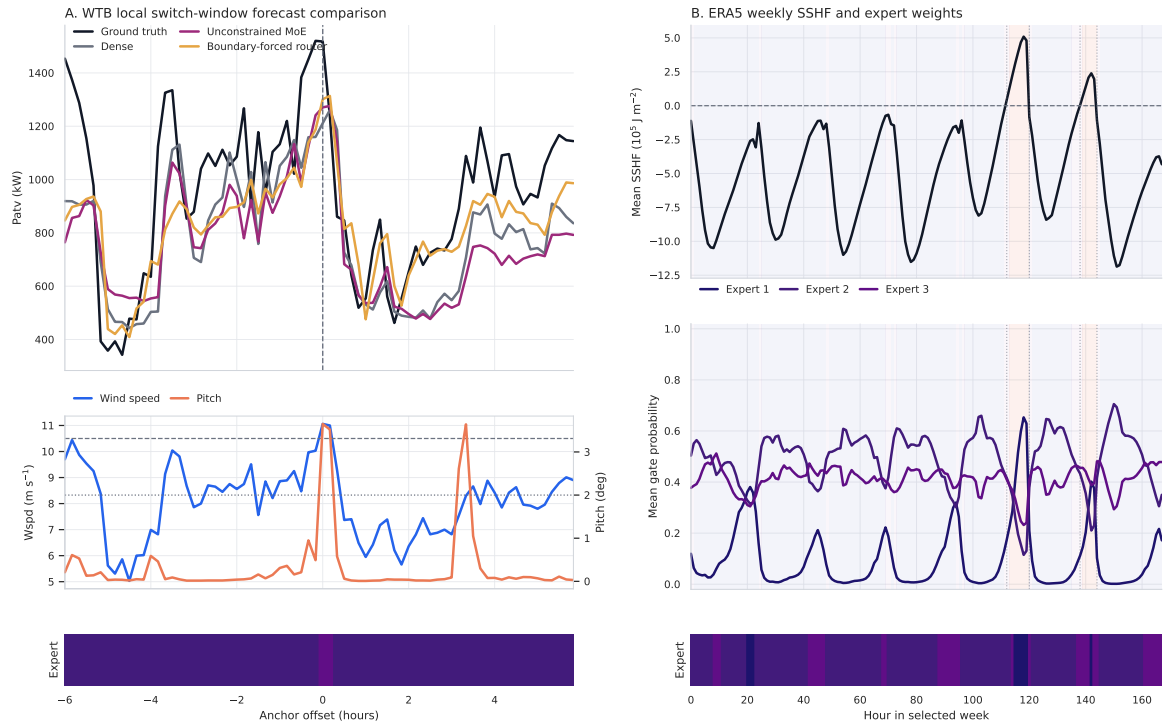


Figure 5: Dual case studies for correction and emergence. Panel A shows a local WTB switch window in which the boundary-forced routing variant remains stable while the dense and unconstrained models deviate more strongly near the MPPT-to-pitch transition; wind speed, pitch angle, and the dominant expert strip are plotted underneath. Panel B shows the ERA5 positive-control view, where sensible heat flux, regime shading, and expert weights evolve coherently over a selected week.

Table. Anchor-only and rule-based router comparison on strict-mask WTB.

Model	Overall	Pitch	NMI/ARI	Noise degrade	Reading
MoE + L_bal + L_align + L_force	236.13	286.95	0.8716/0.9166	5.78	trainable responsibility and reserve coupling
Anchor-only router	233.46	317.81	0.8895/0.9219	105.56	strong semantic rule baseline
Context-supervised router	242.75	287.75	0.7521/0.8020	NA	accurate but less boundary-accountable control
Physics-Aligned MoE	241.42	297.02	0.8312/0.8820	NA	routed forecasting compromise

Table. Energy-system interpretation of shortage-to-reserve cost ratios.

Ratio	System assumption	Economic mapping	Use in paper
2	low scarcity or curtailment-tolerant screening	if reserve costs C_r per MWh, shortage is charged $2 * C_r$	diagnostic only; reserve quantile stays near the median
5	moderate reliability preference for transition-window scheduling	if reserve costs C_r per MWh, shortage is charged $5 * C_r$	usable same-model boundary tradeoff if violation and shortage fall
10	main operating case: shortage materially more expensive than reserve	if reserve costs C_r per MWh, shortage is charged $10 * C_r$	primary reserve diagnostic; not a market-clearing claim
20	scarcity-aware dispatch or high imbalance penalty	if reserve costs C_r per MWh, shortage is charged $20 * C_r$	narrow cost-only region; requires ramp-specific checks
50	emergency reliability or VOLL-like shortage avoidance	if reserve costs C_r per MWh, shortage is charged $50 * C_r$	gate-bin reserve not supported; global policy is safer

5.4 Transition-Window Reserve Diagnostic

Gate-conditioned reserve binning improves the boundary-window cost-violation-shortage tradeoff at a measured reserve-energy price. Four policies isolate the source of the effect: Graph WaveNet/global is the low-RMSE system reference; Graph WaveNet/physical-bin tests physical stratification without a learned gate; boundary router/global isolates the routed model’s forecast penalty with a flat reserve rule; and boundary router/gate-bin adds gate-conditioned allocation to the same routed predictor. The valid attribution is therefore same-model and local: does the learned gate improve the boundary router’s reserve tradeoff near the MPPT-to-pitch transition?

At shortage-to-reserve cost ratio 10, the answer is yes for boundary anchors. Boundary router/gate-bin lowers same-model total cost from 88.13M to 84.58M, violation from 10.38% to 9.00%, and shortage energy from 3.73M to 3.19M normalized units, while reserve energy rises from 50.82M to 52.67M. In percentage terms, shortage energy falls by about 14.5% and reserve energy rises by about 3.6%. Graph WaveNet/global remains the full-sample system reference; the gate-bin effect is a transition-window reserve diagnostic.

Table. Dispatch/reserve main table at shortage-to-reserve cost ratio 10.

Subset	Policy	RMSE penalty	Q	Cost	Violation	Reserve	Shortage	Reading
boundary	Graph WaveNet/physical-bin	+0.00	low=0.900, mid=0.900, high=0.900	84.31M	0.0931	50.35M	3.40M	physics-stratified graph reference
boundary	Boundary router/gate-bin	+10.39	low=0.900, mid=0.900, high=0.900	84.58M	0.0900	52.67M	3.19M	wins on risk, pays reserve
boundary	Boundary router/global	+10.39	all=0.900	88.13M	0.1038	50.82M	3.73M	same-router global reference
boundary	Graph WaveNet/global	+0.00	all=0.900	88.80M	0.1022	50.32M	3.85M	system reference
full	Graph WaveNet/global	+0.00	all=0.900	464.07M	0.0901	288.01M	17.61M	system reference
full	Graph WaveNet/physical-bin	+0.00	low=0.900, mid=0.900, high=0.900	464.43M	0.1057	278.15M	18.63M	physics-stratified graph reference
full	Boundary router/gate-bin	+10.39	low=0.900, mid=0.900, high=0.900	481.36M	0.1214	291.16M	19.02M	cost-only tradeoff
full	Boundary router/global	+10.39	all=0.900	501.32M	0.0857	324.97M	17.64M	same-router global reference

The cost-ratio sweep defines the operating envelope. At ratio 2, the selected gate-bin reserve quantile tolerates too much shortage. At ratios 5–10, gate-bin allocation reduces boundary shortage and violation while carrying more reserve, which is the intended transition-window scheduling regime. At ratio 20 the gain narrows, and at ratio 50 the same-model global rule is safer and cheaper. The gate therefore functions as a conditional reserve-allocation signal for moderate shortage-to-reserve tradeoffs.

The held-out coverage and sensitivity tables explain where the reserve result is reliable. At ratio 10, boundary router/gate-bin is closest to the intended coverage on boundary anchors, with realized coverage 0.910 and violation 0.0900. Horizon and time-of-day slices show that residual shortage is largest at late lead times and in the daytime anchor-index proxy. A horizon-specific empirical-quantile what-if lowers residual shortage in both Graph WaveNet and routed rows, indicating that lead-time calibration matters and that future comparisons should include trained quantile or distributional reserve baselines.

The same evidence can be read as a system-value envelope. The usable interval is the moderate cost-ratio range 5–10, where fewer boundary violations and less shortage are bought with finite additional reserve. This envelope is narrower than the model’s routing success, which is important for practice:

Table. Held-out coverage check for validation-selected reserve quantiles at shortage-to-reserve cost ratio 10.

Subset	Policy	Target cov.	Realized cov.	Error	Violation	Shortage	Reading
boundary	Boundary-forced router/gate-bin	0.900	0.910	+0.010	0.0900	3.19M	near calibration target
boundary	Boundary-forced router/global	0.900	0.896	-0.004	0.1038	3.73M	near calibration target
boundary	Graph WaveNet/global	0.900	0.898	-0.002	0.1022	3.85M	near calibration target
boundary	Graph WaveNet/physical-bin	0.900	0.907	+0.007	0.0931	3.40M	near calibration target
full	Boundary-forced router/gate-bin	0.900	0.879	-0.021	0.1214	19.02M	under-covers on held-out test
full	Boundary-forced router/global	0.900	0.914	+0.014	0.0857	17.64M	near calibration target
full	Graph WaveNet/global	0.900	0.910	+0.010	0.0901	17.61M	near calibration target
full	Graph WaveNet/physical-bin	0.900	0.894	-0.006	0.1057	18.63M	near calibration target

Table. Post-hoc horizon and time-of-day reserve sensitivity for the boundary router/gate-bin policy at shortage-to-reserve cost ratio 10.

Slice	Bucket	Policy	Cost	Violation	Reserve	Shortage	Cells
horizon	h01_h06	Boundary router/gate-bin	21.62M	0.0411	18.53M	0.31M	353950
horizon	h07_h12	Boundary router/gate-bin	24.20M	0.0785	18.47M	0.57M	352927
horizon	h13_h24	Boundary router/gate-bin	65.69M	0.1422	36.28M	2.94M	693326
time_of_day	tod_00_06	Boundary router/gate-bin	44.88M	0.0967	29.21M	1.57M	563849
time_of_day	tod_06_12	Boundary router/gate-bin	26.08M	0.1085	17.22M	0.89M	328791
time_of_day	tod_12_18	Boundary router/gate-bin	16.19M	0.1807	8.52M	0.77M	160400
time_of_day	tod_18_24	Boundary router/gate-bin	24.37M	0.0623	18.32M	0.60M	347163

a gate can be physically meaningful without being the right downstream reserve rule under every cost assumption.

Table. Boundary-window reserve value envelope for the gate-bin policy. Positive reserve means additional reserve energy relative to the same routed model with a global reserve rule.

Ratio	Use window	Cost vs global	Violation vs global	Extra reserve	Shortage vs global	Vs graph bin	System reading
2	not reserve-active	+0.00M	+0.0000	+0.00M	+0.00M	+1.21M	Shortage penalty is too low to activate reserve.
5	usable moderate-cost window	-1.95M	-0.0170	+2.17M	-0.82M	-0.12M	Boundary gate is useful only when fewer violations/shortages justify extra reserve.
10	usable moderate-cost window	-3.55M	-0.0138	+1.85M	-0.54M	+0.26M	Boundary gate is useful only when fewer violations/shortages justify extra reserve.
20	narrow cost-only window	-1.39M	+0.0001	-0.70M	-0.03M	+1.49M	Benefit narrows; use only after ramp-specific checks.
50	not applicable at high penalty	+5.93M	+0.0035	-1.04M	+0.14M	+6.95M	Same-model global policy is safer at this penalty.

Operator-facing slices sharpen the same point. Gate-bin reserve helps the MPPT-to-pitch and low-ramp slices, while pitch-to-MPPT and high-ramp slices favor other reserve choices. The operational decision curve therefore places the routed model’s RMSE penalty beside the reserve-risk benefit: the higher-RMSE router is considered where fewer boundary shortages and violations are worth the reserve-energy cost.

The three operational cases in the next table show how routing quality and downstream forecast value can diverge. A correct gate can help allocate attention around the MPPT-to-pitch boundary, while a poor downstream forecast or an unsuitable reserve bin can still make the operator reject the gate-bin rule.

The paired uncertainty table is intentionally conservative. It compares full-sample reserve outcomes against Graph WaveNet/global using seed-paired differences with bootstrap intervals and a sign-permutation check. These rows do not show a stable whole-sample cost advantage for the routed policies; the intervals are wide with five seeds. The reserve benefit is therefore the boundary-window operational pattern above, while the whole-sample comparison remains a diagnostic check.

Lead-lag and high-error audits define the limit of this reserve interpretation. The corrected gate aligns most strongly at and after the physical switch, not several steps before it, so the gate is a state diagnostic rather than a lead-time alarm. High-error cases show the complementary failure mode: the gate can assign the correct operating regime while the expert forecast remains numerically poor. Correct routing is therefore necessary for mechanism interpretation, but it is not sufficient for accurate power prediction.

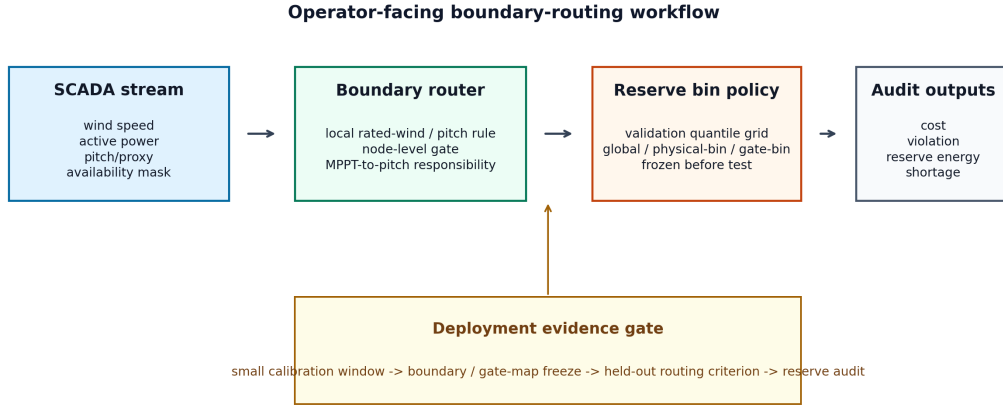


Figure 6: Operator-facing workflow for the reserve audit. SCADA fields define the local operating boundary, the boundary router assigns MPPT-to-pitch responsibility, the reserve-bin policy is calibrated on validation shortfall and frozen before test, and the reported outputs are cost, violation, reserve energy, and shortage energy.

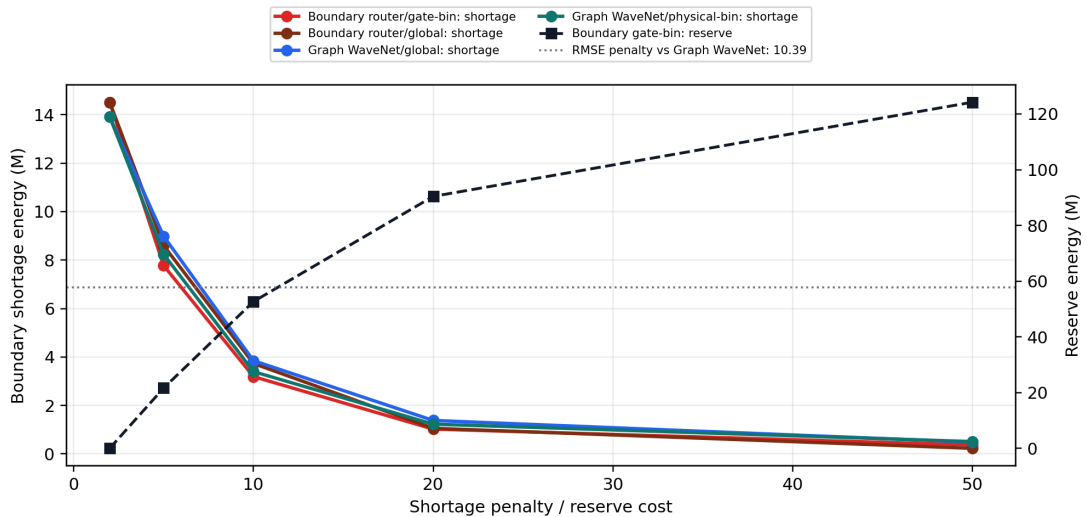


Figure 7: Operational decision curve for the WTB boundary window. The chart places the boundary router's RMSE penalty against Graph WaveNet/global beside reserve energy and shortage-energy changes across shortage-to-reserve cost ratios. The figure explains the restricted decision logic: the routed model is not chosen for average error, but examined for a boundary-window reserve tradeoff that is acceptable only in the moderate cost-ratio envelope.

Table. Horizon-specific empirical-quantile what-if. This recalibrates empirical shortfall quantiles by horizon bucket without training a probabilistic forecaster.

Subset	Policy	Pooled cost	Horizon-Q cost	Delta violation	Delta reserve	Delta shortage	Reading
boundary	Boundary router/gate-bin	111.51M	107.03M	-0.0045	+0.79M	-0.53M	risk improves, post-hoc only
boundary	Boundary router/global	112.12M	107.74M	-0.0055	+1.24M	-0.56M	risk improves, post-hoc only
boundary	Graph WaveNet/global	111.27M	106.76M	-0.0054	+0.91M	-0.54M	risk improves, post-hoc only
boundary	Graph WaveNet/physical-bin	109.83M	104.27M	-0.0032	-0.36M	-0.52M	risk improves, post-hoc only
full	Boundary router/gate-bin	516.49M	494.01M	-0.0002	-13.17M	-0.93M	risk improves, post-hoc only
full	Boundary router/global	531.00M	514.86M	-0.0042	-7.39M	-0.88M	risk improves, post-hoc only
full	Graph WaveNet/global	492.78M	469.99M	-0.0056	-15.97M	-0.68M	risk improves, post-hoc only
full	Graph WaveNet/physical-bin	487.75M	460.81M	-0.0040	-20.09M	-0.68M	risk improves, post-hoc only

Table. Operator-facing cases: when the boundary gate helps and when the downstream decision rejects it.

Case	Observed	Operator reading	Boundary reading
MPTT-to-pitch reserve	global cost 16.33M, violation 0.1124, shortage 0.77M, gate-bin cost 17.93M, violation 0.1027, shortage 0.69M	gate-bin lowers transition-window violation and shortage but costs more reserve	boundary-window diagnostic, not global reserve optimization
high-ramp reserve	global violation 0.0766, shortage 1.06M, reserve 57.03M; gate-bin violation 0.1253, shortage 2.51M, reserve 42.02M	gate-bin can look cheaper by carrying too little reserve	failure slice, require ramp-specific checks
gate-correct but forecast-bad	model MoE + L _{bul} + L _{align} + L _{force} , seed 203, gate_correct 1, RMSE 1258.50	right responsibility assignment does not guarantee an accurate value map	mechanism accountability, not dispatch-ready point forecasting

5.5 Stress Tests and Ablation Frontier

Routing semantics transfer to unseen turbines within WTB. Holding out 27 of 134 turbines during training and validation, the boundary-forced router reaches NMI/ARI 0.8340 +/- 0.0958 / 0.8829 +/- 0.0875 on the held-out nodes, with overall RMSE 233.87 +/- 2.84. A pre-specified later WTB holdout gives a similar temporal stress result: NMI/ARI 0.8352 +/- 0.0669 / 0.8870 +/- 0.0669 and overall RMSE 235.49 +/- 7.19. These tests support within-farm spatial and temporal robustness of routing semantics, while leaving value-prediction stability and new-farm transfer to separate checks.

The WTB proxy deployment drill tests whether the calibration workflow can be executed under held-out separation. A 2-day calibration window followed by future-period testing passes the held-out routing gate in all five seeds, with mean NMI/ARI 0.8397/0.8907 and balanced accuracy 0.9770. A 7-day east-turbine proxy also passes in all five seeds, with mean NMI/ARI 0.8272/0.8781 and balanced accuracy 0.9709. These WTB-internal demonstrations show that the workflow can be executed with held-out support before a site-level reserve claim is made.

The drill also defines practical usability checks. Thresholds must be chosen on a pre-specified local grid and frozen before held-out testing. Calibration support depends on valid boundary cells and pitch observability, not elapsed days alone. If pitch channels or validated proxies are missing, the gate cannot be interpreted as a physical router; if high-ramp slices fail the reserve audit, the gate-bin rule should not be used for reserve allocation.

The ablation exposes a controllable accuracy-semantics frontier. Unconstrained MoE sits in the low-semantics region. Regime alignment moves the gate sharply upward in NMI/ARI with modest RMSE cost, while boundary forcing pushes further toward semantic repair at a clearer forecasting price. With load balancing, regime alignment, and boundary forcing active, the selected corrected router reaches NMI/ARI 0.8716 +/- 0.0418 / 0.9166 +/- 0.0371. The ablation claim is therefore the shape of the RMSE/NMI frontier, not an accuracy ranking.

The mechanism and fairness checks rule out two easy explanations for the WTB result. The in-family routed variants share model capacity, and the external baselines show that the routed model is not winning by hidden parameter count or speed. The compute-cost table adds the practical constraint: PatchTST is fastest, Graph WaveNet has the lowest strict-mask RMSE, and the routed models sit between them with about 110k trainable parameters and 13–14 s evaluation time for the strict test windows.

The intervention audit shows that the gate depends selectively on the intended boundary anchor. Zeroing the boundary anchor raises overall RMSE by 2.6792 and drops NMI/ARI by 0.7017 / 0.8593; zeroing the wake score is near-null, with overall RMSE change 0.0031 and NMI/ARI drop about 0.0010 / 0.0010. Placebo labels built from temporal shifts, node permutation, within-time shuffling, and global shuffling do not reproduce the actual-label agreement. These checks move the WTB claim beyond visualization:

Table. Boundary-window operational reserve slices at shortage-to-reserve cost ratio 10.

Slice	Policy	Cost	Violation	Reserve	Shortage	Valid cells
mppt_to_pitch	Boundary-forced router/global	16.33M	0.1124	8.62M	0.77M	1968
mppt_to_pitch	Boundary-forced router/physical-bin	16.36M	0.0936	10.72M	0.56M	1968
mppt_to_pitch	Graph WaveNet/physical-bin	17.43M	0.0981	10.83M	0.66M	1968
mppt_to_pitch	Graph WaveNet/global	17.55M	0.1109	9.33M	0.82M	1968
mppt_to_pitch	Boundary-forced router/gate-bin	17.93M	0.1027	11.02M	0.69M	1968
pitch_to_mppt	Boundary-forced router/global	20.27M	0.0604	17.38M	0.29M	1992
pitch_to_mppt	Boundary-forced router/physical-bin	20.65M	0.0772	17.16M	0.35M	1992
pitch_to_mppt	Boundary-forced router/gate-bin	20.83M	0.0833	16.72M	0.41M	1992
pitch_to_mppt	Graph WaveNet/global	21.26M	0.0982	15.49M	0.58M	1992
pitch_to_mppt	Graph WaveNet/physical-bin	21.59M	0.1047	15.21M	0.64M	1992
high_ramp	Graph WaveNet/physical-bin	65.83M	0.1224	45.76M	2.01M	6212
high_ramp	Boundary-forced router/gate-bin	67.08M	0.1253	42.02M	2.51M	6212
high_ramp	Graph WaveNet/global	67.15M	0.0965	51.67M	1.55M	6212
high_ramp	Boundary-forced router/global	67.67M	0.0766	57.03M	1.06M	6212
high_ramp	Boundary-forced router/physical-bin	68.02M	0.1065	49.96M	1.81M	6212
low_ramp	Boundary-forced router/gate-bin	10.73M	0.0845	6.70M	0.40M	5594
low_ramp	Graph WaveNet/physical-bin	10.84M	0.0822	6.85M	0.40M	5594
low_ramp	Graph WaveNet/global	10.88M	0.0817	6.95M	0.39M	5594
low_ramp	Boundary-forced router/physical-bin	11.15M	0.0837	6.79M	0.44M	5594
low_ramp	Boundary-forced router/global	11.24M	0.0849	6.84M	0.44M	5594

the gate is tied to the MPPT-to-pitch boundary rather than to arbitrary physical channels or shuffled label structure.

The time-forward audit gives the main failure case. Routing semantics remain stable in the late test block, with NMI/ARI 0.8661 +/- 0.0393 / 0.9079 +/- 0.0364, but forecasting accuracy worsens sharply: late-minus-early overall RMSE is +50.3454 and switch-window RMSE is +63.3549. The distribution-shift diagnostics show higher target power, larger pitch action, lower mask-valid coverage, and shifted regime shares. The gate still tracks the physical partition, but the value map faces a different operating distribution. This is a useful boundary condition: the method supports routing semantics more strongly than time-stable forecasting accuracy.

Sensitivity checks show that the routing result survives nearby threshold and weight changes. Varying λ_{align} from 2500 to 7500 keeps NMI in the 0.8255–0.8546 range, and varying λ_{force} from 5000 to 15000 keeps NMI in the 0.8091–0.8406 range. Re-labeling saved strict-mask gate outputs over the local rated-wind/pitch grid gives worst-case NMI/ARI 0.8655 +/- 0.0429 / 0.9146 +/- 0.0377 with minimum shared-valid support 0.9909. Within that pre-specified WTB grid, the operating-boundary rule remains stable.

5.6 External Failure and Local Deployment Protocol

Kelmarsh/Penmanshiel external testing fails the pre-specified routing criterion, and this negative result is part of the contribution. The default WTB boundary gives mean external routing NMI 0.4877, below the 0.50 criterion. Validation-only local recalibration raises default test NMI only from 0.1324 to 0.1491 on average, and the small calibration-window adaptation remains below the held-out balanced-accuracy threshold. The failure is therefore not a simple threshold shift. It points to concrete deployment conditions: pitch/proxy coverage, boundary-cell support, turbine geometry, farm scale, power-curve distribution, sensor fields, and regime-label balance.

This external failure turns into a local evidence protocol. A new farm must first show pitch observability or a validated pitch proxy; then re-estimate rated wind, pitch threshold, boundary band, and gate-to-regime map on a local calibration period; and only then pass a held-out routing criterion before any

Table. Boundary-window cost-ratio sensitivity in reserve-system language.

Ratio	Policy	Q	Cost	Violation	Reserve	Shortage	Vs global
2	Boundary router/gate-bin	low=0.500, mid=0.500, high=0.500	29.02M	0.3420	0.00M	14.51M	not a win
2	Boundary router/global	all=0.500	29.02M	0.3420	0.00M	14.51M	reference
2	Graph WaveNet/global	all=0.500	27.82M	0.3237	0.00M	13.91M	reference
2	Graph WaveNet/physical-bin	low=0.500, mid=0.500, high=0.500	27.82M	0.3237	0.00M	13.91M	not a win
5	Boundary router/gate-bin	low=0.800, mid=0.500, high=0.800	60.50M	0.2029	21.62M	7.77M	wins on risk, pays reserve
5	Boundary router/global	all=0.800	62.44M	0.2199	19.45M	8.60M	reference
5	Graph WaveNet/global	all=0.800	62.61M	0.2255	17.69M	8.98M	reference
5	Graph WaveNet/physical-bin	low=0.800, mid=0.800, high=0.800	60.62M	0.2043	19.42M	8.24M	wins on risk, pays reserve
10	Boundary router/gate-bin	low=0.900, mid=0.900, high=0.900	84.58M	0.0900	52.67M	3.19M	wins on risk, pays reserve
10	Boundary router/global	all=0.900	88.13M	0.1038	50.82M	3.73M	reference
10	Graph WaveNet/global	all=0.900	88.80M	0.1022	50.32M	3.85M	reference
10	Graph WaveNet/physical-bin	low=0.900, mid=0.900, high=0.900	84.31M	0.0931	50.35M	3.40M	wins on risk, pays reserve
20	Boundary router/gate-bin	low=0.950, mid=0.950, high=0.950	110.90M	0.0375	90.41M	1.02M	cost-only tradeoff
20	Boundary router/global	all=0.950	112.30M	0.0374	91.11M	1.06M	reference
20	Graph WaveNet/global	all=0.950	113.09M	0.0443	85.44M	1.38M	reference
20	Graph WaveNet/physical-bin	low=0.950, mid=0.950, high=0.950	109.41M	0.0411	84.84M	1.23M	operational win
50	Boundary router/gate-bin	low=0.975, mid=0.975, high=0.975	142.73M	0.0160	124.20M	0.37M	risk loss
50	Boundary router/global	all=0.975	136.80M	0.0125	125.24M	0.23M	reference
50	Graph WaveNet/global	all=0.975	136.90M	0.0236	112.61M	0.49M	reference
50	Graph WaveNet/physical-bin	low=0.975, mid=0.975, high=0.975	135.78M	0.0234	110.26M	0.51M	cost-only tradeoff

Table. Paired uncertainty for full-sample reserve metrics versus Graph WaveNet/global.

Candidate	Metric	Delta	95% CI	P(lower)	Perm p
Graph WaveNet/physical-bin	total_cost	+0.36M	[-1.14M, +2.28M]	0.400	0.810
Graph WaveNet/physical-bin	violation_rate	+0.0156	[+0.0143, +0.0168]	0.000	0.065
Graph WaveNet/physical-bin	reserve_energy	-9.87M	[-10.86M, -8.83M]	1.000	0.065
Graph WaveNet/physical-bin	shortage_energy	+1.02M	[+0.80M, +1.28M]	0.000	0.065
Physics-Aligned MoE/gate-bin	total_cost	+30.67M	[-63.36M, +111.49M]	0.400	0.629
Physics-Aligned MoE/gate-bin	violation_rate	+0.0251	[+0.0098, +0.0391]	0.200	0.127
Physics-Aligned MoE/gate-bin	reserve_energy	+9.07M	[-14.62M, +30.47M]	0.200	0.498
Physics-Aligned MoE/gate-bin	shortage_energy	+2.16M	[-5.94M, +9.57M]	0.400	0.685
Boundary-forced router/global	total_cost	+37.25M	[-31.51M, +107.97M]	0.400	0.380
Boundary-forced router/global	violation_rate	-0.0044	[-0.0198, +0.0122]	0.600	0.631
Boundary-forced router/global	reserve_energy	+36.95M	[+21.02M, +56.35M]	0.000	0.065
Boundary-forced router/global	shortage_energy	+0.03M	[-5.77M, +6.74M]	0.600	1.000
Boundary-forced router/gate-bin	total_cost	+17.28M	[-52.09M, +87.95M]	0.600	0.752
Boundary-forced router/gate-bin	violation_rate	+0.0312	[+0.0151, +0.0529]	0.000	0.065
Boundary-forced router/gate-bin	reserve_energy	+3.15M	[-10.24M, +17.86M]	0.400	0.628
Boundary-forced router/gate-bin	shortage_energy	+1.41M	[-4.59M, +8.42M]	0.600	0.752

gate-bin reserve rule is used. If the held-out gate fails, the result remains a boundary-condition diagnosis rather than a physical-router or reserve-control claim.

The practical rule is now clear. Correction strength should scale with regime observability and with the operating decision that consumes the gate. ERA5 already exposes its regime marker, so routing correction improves alignment without changing the headline accuracy story. WTB hides part of the MPPT-to-pitch boundary inside control action, so stronger gate repair is justified when its reserve-window diagnostic value exceeds the measured RMSE and reserve-energy cost. The method is most appropriate when the operating boundary is part of the scientific or operational question.

6 Limitations

The WTB correction depends on threshold-based pseudo-labels derived from wind speed and mean pitch angle. Those same channels also appear in the gate anchor, so the method deliberately constrains the router to a declared operating boundary. The threshold audit and loss-weight sweep show local robustness, but they do not exhaust all anchor thresholds, wake-score cutoffs, wake-cone parameters, turbine-control

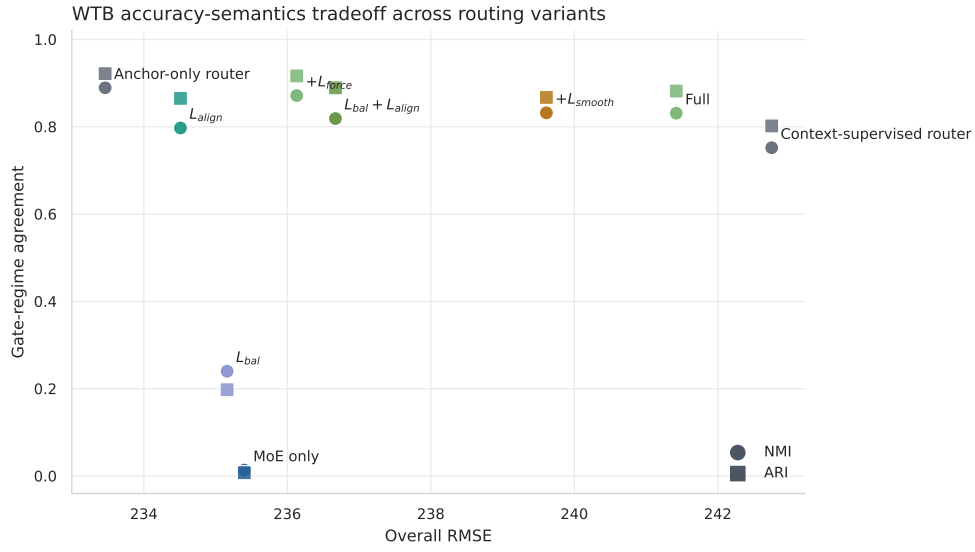


Figure 8: WTB accuracy-semantics tradeoff across routing variants. Circles show NMI and squares show ARI against overall RMSE. The frontier clarifies that stronger routing correction improves gate-regime agreement but can move the model away from the lowest forecast error.

New wind-farm local evidence protocol: physical interpretation requires every check to pass

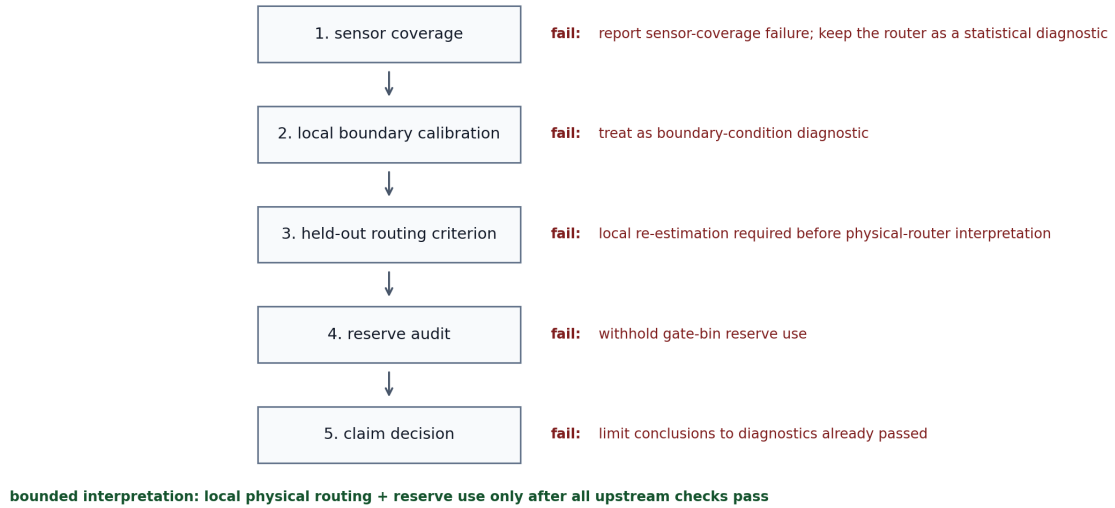


Figure 9: New wind-farm evidence protocol. The protocol moves from sensor coverage to local boundary calibration, held-out routing, reserve audit, and finally bounded operating claims.

Table. Quasi-external deployment drill inside WTB: calibration-only boundary/gate-map selection followed by held-out testing.

Scenario	Calib. days	Runs	Min boundary cells	Held-out NMI/ARI	Bal. acc.	Decision	Boundary reading
future-period WTB proxy	2.0	5/5	67	0.8397/0.8907	0.9770	passes pre-specified held-out routing gate	supports calibration-to-held-out workflow inside WTB; not external-farm portability
east-turbine WTB proxy	7.0	5/5	103	0.8272/0.8781	0.9709	passes pre-specified held-out routing gate	supports calibration-to-held-out workflow inside WTB; not external-farm portability

Table. Parameters, training cost, and inference speed on strict-mask WTB runs.

Model	Params	Train h	Eval s	Win/s	Reading
PatchTST	111512	0.09	4.08	1221.7	fastest inference
Physics-Aligned MoE	110012	0.62	13.24	376.2	router adds interpretable responsibility at moderate cost
Boundary-forced router	110012	0.59	13.89	371.0	router adds interpretable responsibility at moderate cost
Graph WaveNet	123800	1.50	23.01	216.5	lowest strict-mask RMSE, slower

settings, or retrained no-/lagged-anchor ablations. The method is boundary-aware regularized routing, not fully learned operating-state discovery.

The active-power anchor has a strict timing requirement. Patv is allowed only as the historical/anchor-time active-power measurement available when the forecast is issued; future active power remains the supervised target. This is a defensible SCADA forecasting convention, but it is also a portability constraint and a possible source of interpretive circularity. The current intervention evidence zeroes the boundary-anchor input and shows that routing agreement collapses, but it is not a retrained no-Patv or lagged-Patv ablation. Taken together with the missing wind-speed and pitch ablations above, the present evidence cannot claim to identify which part of the gate is pitch/wind observability, active-power status, or structure learned after anchoring. If a deployment environment delays active-power telemetry, changes channel definitions, or evaluates a decision before the current active-power value is available, the Patv anchor must be removed or lagged and the leakage, intervention, and reserve diagnostics must be rerun.

The expert-regime language is tied to the implemented logit convention. The first primary-regime logits are fixed to the operating labels during supervised alignment, which prevents seed-wise permutation for those anchored meanings. That makes cross-seed MPPT/pitch statements interpretable, but only for the anchored logits and only under the declared mapping. Unassigned experts and wake auxiliary logits should not be overread as universal turbine states.

The comparison is bounded. WTB includes graph, transformer, lag-feature, persistence, power-curve, and DLinear-style references, while ERA5 includes persistence and several learned baselines. Larger time-series backbones, broader graph-transformer variants, trained quantile-regression baselines, and scenario/distributional reserve methods are outside the present benchmark.

The reserve audit is deliberately narrower than power-system dispatch. It does not implement unit commitment, market clearing, a full probabilistic prediction-to-reserve pipeline, real dispatch integration, or a closed-loop operator simulation. The audit is therefore a normalized proxy decision: fixed empirical quantile bins are calibrated on validation shortfall and tested once on held-out predictions, with costs reported in normalized reserve-energy units rather than money. It is useful for ranking boundary-window risk tradeoffs under declared cost ratios, but it is not a market-price, delivery-constrained, or security-constrained dispatch study.

External wind-farm diagnostics fail the external-site routing criterion. The quasi-external drill shows that the calibration-to-held-out workflow can pass inside WTB, but it does not change the Kelmarsh/Penmanshiel conclusion. The evidence supports within-WTB spatial and temporal stress validation plus external diagnosis of when local boundary re-estimation is required.

The time-forward audit is not an external validation. It is a post-hoc split of saved WTB test predictions into contiguous anchor-time blocks. It shows that the strict-mask gate keeps its physical partition in the late block, but it also shows a large late-test RMSE increase. The failure case is therefore explicit: late-period distribution shift preserves routing agreement while degrading value prediction. This

Table 3. Replay-safe strict-mask WTB mechanism interventions.

Intervention	Delta RMSE	Delta switch RMSE	Drop NMI	Drop ARI
anchor_boundary_zero	2.6792 [0.7347, 4.5598]	2.9470 [0.7985, 5.1213]	0.7017 [0.6451, 0.7911]	0.8593 [0.8208, 0.9075]
anchor_wake_zero	0.0031 [-0.0016, 0.0110]	0.0032 [-0.0011, 0.0106]	0.0010 [0.0001, 0.0021]	0.0010 [0.0002, 0.0023]
anchor_all_zero	7.2037 [2.1205, 12.2869]	7.7592 [2.2208, 13.2587]	0.7191 [0.6579, 0.8230]	0.8638 [0.8244, 0.9179]

Table 4. Strict-mask WTB routing placebo separations.

Control	Delta NMI	Delta ARI
temporal_shift_18	0.5875 [0.5551, 0.6210]	0.5759 [0.5466, 0.6043]
temporal_shift_144	0.8324 [0.8020, 0.8692]	0.8656 [0.8345, 0.8954]
node_permutation	0.3289 [0.3016, 0.3659]	0.2825 [0.2581, 0.3100]
within_time_shuffle	0.3337 [0.3040, 0.3702]	0.2872 [0.2605, 0.3153]
global_shuffle	0.8716 [0.8411, 0.9086]	0.9166 [0.8879, 0.9422]

is an important boundary condition: the method currently supports routing semantics, not time-stable forecasting accuracy.

Statistical reliability is uneven across claim types. The strongest claims are the five-seed WTB routing-recovery, mechanism-intervention, and within-WTB spatial/temporal stress results. Engineering base-lines, reserve-window diagnostics, and external adaptation checks are reported as deterministic or bounded diagnostics, not as broad superiority claims. WTB and ERA5 are sufficient for the intended weak-versus-visible observability contrast, but other industrial or geophysical systems require local boundary estimation and held-out routing checks.

7 Conclusion

This paper treats non-stationary wind-power modeling as a routing-interpretability problem around operating transitions. Strong graph, transformer, and lag-feature baselines remain better mean-error forecasters than the boundary-forced router, but the selected router recovers the WTB MPPT-to-pitch partition, survives placebo and within-WTB spatial/temporal stress checks, and links that partition to a transition-window reserve diagnostic. At shortage-to-reserve cost ratio 10, gate-conditioned reserve binning reduces same-model boundary-window violation and shortage energy at a measured reserve-energy cost.

The practical conclusion is bounded. When the regime marker is visible, routing regularization calibrates an already available partition. When the marker is partly hidden by turbine-control action, physics guidance at the gate can repair a weakly identified partition and provide reserve/ramp-risk diagnosis, but it trades against forecast error and does not create a whole-sample reserve rule. Kelmarsh/Penmanshiel failure shows that this responsibility claim must be re-established when turbine geometry, pitch observability, sensor fields, power curves, or label distributions change. Until a new wind farm passes the local evidence protocol, the method should be described as within-WTB operating-boundary interpretability plus external failure-boundary diagnosis plus transition-window reserve diagnosis.

8 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used OpenAI ChatGPT/Codex to support language editing, consistency checking, and submission-material drafting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published

Table 5. Strict-mask WTB late-period distribution-shift diagnostics.

Model	Delta RMSE	Delta abs ramp	Regime TV	Regime JSD	Delta NMI
MoE + L_bal + L_align + L_force	50.3454 [46.8137, 54.1758]	-1104.6378 [-1104.6378, -1104.6378]	0.2983	0.0539	0.0068 [0.0021, 0.0127]

Table. WTB threshold/label validity audit for saved strict-mask gate outputs.

Rated wind	Pitch threshold	Runs	NMI		ARI	Shared valid
10.0	1.5	5	0.8724 ± 0.0404	0.9165 ± 0.0368		0.9956
10.0	2.0	5	0.8721 ± 0.0407	0.9164 ± 0.0369		0.9991
10.0	2.5	5	0.8717 ± 0.0410	0.9163 ± 0.0370		0.9990
10.5	1.5	5	0.8723 ± 0.0418	0.9165 ± 0.0372		0.9960
10.5	2.0	5	0.8710 ± 0.0420	0.9161 ± 0.0374		1.0000
10.5	2.5	5	0.8696 ± 0.0422	0.9157 ± 0.0374		0.9998
11.0	1.5	5	0.8687 ± 0.0426	0.9156 ± 0.0375		0.9909
11.0	2.0	5	0.8672 ± 0.0429	0.9151 ± 0.0377		0.9949
11.0	2.5	5	0.8655 ± 0.0429	0.9146 ± 0.0377		0.9949

article.

9 Code and data availability

The raw datasets used in this study are publicly available from their original providers: the KDD Cup 2022 wind-farm SCADA benchmark, ECMWF ERA5 reanalysis, and the Kelmarsh and Penmanshiel wind-farm SCADA records. Raw third-party data are not redistributed in the manuscript package and remain subject to the terms of the source providers. Code, configuration files, releasable derived tables and figure data, model checkpoints where licensing permits, and reproduction instructions for the WTB routing analysis, transition-window reserve audit, quasi-external WTB drill, and external boundary diagnostics will be made available with the article. The analysis uses turbine and atmospheric measurements only and involves no human participants or human-subject data.

Table. Why external wind-farm boundary transfer failed and how the result is used.

Evidence gate	Observed	Decision	Bounded claim
pre-specified external routing criterion	80/80 runs complete; mean NMI 0.4877 below threshold 0.50; mean ARI 0.5112	failed external-site routing criterion	negative boundary-condition evidence only
validation-selected local boundary recalibration	default test NMI 0.1324 -> recalibrated 0.1491 (delta +0.0167)	small average recovery, not portability	local boundary re-estimation is required
small calibration-window gate-map adaptation	40/40 routing runs adapted; chronological balanced accuracy 0.4787 below 0.50	completed negative adaptation diagnostic	calibration window is insufficient without held-out pass
sensor and boundary support	Penmanshiel-to-Kelmarsh leave-one pitch-feature coverage 0.0000 and effective boundary cells 0	label not observable in this transfer direction	external physical-router interpretation is not supported
chronological-only local recalibration	boundary router chronological NMI recovery +0.0311; held-out test NMI after recalibration 0.2372	partial local recovery but still weak	generalization requires local held-out checks
single-site apparent improvement	Penmanshiel chronological adapted balanced accuracy 0.5000, but NMI remains 0.0000	not enough for a positive external claim	future work must satisfy a held-out criterion before site-level use

References

- Bai, L., Yao, L., Li, C., Wang, X., Wang, C., 2020. Adaptive graph convolutional recurrent network for traffic forecasting, in: *Advances in Neural Information Processing Systems*. pp. 17804–17815.
- Bremnes, J.B., 2004. Probabilistic wind power forecasts using local quantile regression. *Wind Energy* 7, 47–54. <https://doi.org/10.1002/we.107>
- Cao, Y., Wang, J., 2026. ECTO: Exogenous-conditioned temporal operator for ultra-short-term wind power forecasting.
- Daenens, S., Verstraeten, T., Daems, P.-J., Nowé, A., Helsen, J., 2025. Spatio-temporal graph neural networks for power prediction in offshore wind farms using SCADA data. *Wind Energy Science* 10, 1137–1152. <https://doi.org/10.5194/wes-10-1137-2025>
- Doherty, R., O'Malley, M., 2005. A new approach to quantify reserve demand in systems with significant installed wind capacity. *IEEE Transactions on Power Systems* 20, 587–595. <https://doi.org/10.1109/TPWRS.2005.846206>
- Ela, E., Milligan, M., Kirby, B., 2011. Operating reserves and variable generation (No. NREL/TP-5500-51978). National Renewable Energy Laboratory, Golden, CO. <https://doi.org/10.2172/1023095>
- Fedus, W., Zoph, B., Shazeer, N., 2022. Switch transformers: Scaling to trillion parameter models with simple and efficient sparsity. *Journal of Machine Learning Research* 23, 1–39.
- Gallego-Castillo, C., Cuerva-Tejero, A., Lopez-Garcia, O., 2015. A review on the recent history of wind power ramp forecasting. *Renewable and Sustainable Energy Reviews* 52, 1148–1157. <https://doi.org/10.1016/j.rser.2015.07.154>
- Gao, J., Cheng, Y., Zhang, D., Chen, Y., 2025. Physics-constrained wind power forecasting aligned with probability distributions for noise-resilient deep learning. *Applied Energy* 383, 125295. <https://doi.org/10.1016/j.apenergy.2025.125295>
- Guo, S., Lin, Y., Feng, N., Song, C., Wan, H., 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting, in: *Proceedings of the AAAI Conference on Artificial Intelligence*. pp. 922–929. <https://doi.org/10.1609/aaai.v33i01.3301922>
- Haq, I.U., Kumar, A., Rathore, P.S., 2025. Machine learning approaches for wind power forecasting: A comprehensive review. *Discover Applied Sciences* 7, 1139. <https://doi.org/10.1007/s42452-025-07675-x>
- Hersbach, H., Bell, B., Berrisford, P., Hirahara, S., Horányi, A., Muñoz-Sabater, J., Nicolas, J., Peubey, C., Radu, R., Schepers, D., Simmons, A., Soci, C., Abdalla, S., Abellan, X., Balsamo, G., Bechtold, P., Biavati, G., Bidlot, J.-R., Bonavita, M., De Chiara, G., Dahlgren, P., Dee, D., Diamantakis, M., Dragani, R., Flemming, J., Forbes, R., Fuentes, M., Geer, A., Haimberger, L., Healy, S., Hogan, R., Hólm, E., Janisková, M., Keeley, S., Laloyaux, P., Lopez, P., Lupu, C., Radnoti, G., Rosnay, P. de, Rozum, I., Vamborg, F., Villaume, S., Thépaut, J.-N., 2020. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society* 146, 1999–2049. <https://doi.org/10.1002/qj.3803>
- Jacobs, R.A., Jordan, M.I., Nowlan, S.J., Hinton, G.E., 1991. Adaptive mixtures of local experts. *Neural Computation* 3, 79–87. <https://doi.org/10.1162/neco.1991.3.1.79>
- Jordan, M.I., Jacobs, R.A., 1994. Hierarchical mixtures of experts and the EM algorithm. *Neural Computation* 6, 181–214. <https://doi.org/10.1162/neco.1994.6.2.181>
- Karniadakis, G.E., Kevrekidis, I.G., Lu, L., Perdikaris, P., Wang, S., Yang, L., 2021. Physics-informed machine learning. *Nature Reviews Physics* 3, 422–440. <https://doi.org/10.1038/s42254-021-00314-5>
- Karpatne, A., Atluri, G., Faghmous, J.H., Steinbach, M., Banerjee, A., Ganguly, A., Shekhar, S., Samatova, N.F., Kumar, V., 2017. Theory-guided data science: A new paradigm for scientific discovery from data. *IEEE Transactions on Knowledge and Data Engineering* 29, 2318–2331. <https://doi.org/10.1109/TKDE.2017.2720168>
- Kim, D., Ryu, G., Moon, C., 2024. Accuracy of a short-term wind power forecasting model based on deep learning using LiDAR-SCADA integration: A case study of the 400-MW Anholt offshore wind farm. *Applied Energy* 373, 123882. <https://doi.org/10.1016/j.apenergy.2024.123882>

- Li, Y., Yu, R., Shahabi, C., Liu, Y., 2018. Diffusion convolutional recurrent neural network: Data-driven traffic forecasting, in: International Conference on Learning Representations.
- Nielsen, H.A., Madsen, H., Nielsen, T.S., 2006. Using quantile regression to extend an existing wind power forecasting system with probabilistic forecasts. *Wind Energy* 9, 95–108. <https://doi.org/10.1002/we.180>
- Park, J., Park, J., 2019. Physics-induced graph neural network: An application to wind-farm power estimation. *Energy* 187, 115883. <https://doi.org/10.1016/j.energy.2019.115883>
- Parsa, S.M., 2025. Physics-informed machine learning meets renewable energy systems: A review of advances, challenges, guidelines, and future outlooks. *Applied Energy* 402, 126925. <https://doi.org/10.1016/j.apenergy.2025.126925>
- Pinson, P., 2013. Wind energy: Forecasting challenges for its operational management. *Statistical Science* 28, 564–585. <https://doi.org/10.1214/13-STS445>
- Pinson, P., Chevallier, C., Kariniotakis, G.N., 2007. Trading wind generation from short-term probabilistic forecasts of wind power. *IEEE Transactions on Power Systems* 22, 1148–1156. <https://doi.org/10.1109/TPWRS.2007.901117>
- Raissi, M., Perdikaris, P., Karniadakis, G.E., 2019. Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations. *Journal of Computational Physics* 378, 686–707. <https://doi.org/10.1016/j.jcp.2018.10.045>
- Read, J.S., Jia, X., Willard, J., Appling, A.P., Zwart, J.A., Oliver, S.K., Karpatne, A., Hansen, G.J.A., Hanson, P.C., Watkins, W., Steinbach, M., Kumar, V., 2019. Process-guided deep learning predictions of lake water temperature. *Water Resources Research* 55, 9173–9190. <https://doi.org/10.1029/2019WR024922>
- Shazeer, N., Mirhoseini, A., Maziarz, K., Davis, A., Le, Q., Hinton, G., Dean, J., 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer, in: International Conference on Learning Representations.
- Shi, S., Han, L., Liu, Y., others, 2025. Time-MoE: Billion-scale time series foundation models with mixture of experts, in: International Conference on Learning Representations.
- Tautz-Weinert, J., Watson, S.J., 2017. Using SCADA data for wind turbine condition monitoring: A review. *IET Renewable Power Generation* 11, 382–394. <https://doi.org/10.1049/iet-rpg.2016.0248>
- Tian, J., Ding, Y., Xu, R., Miao, H., Guo, C., Yang, B., 2026. ARROW: An adaptive rollout and routing method for global weather forecasting.
- Wang, J., Botterud, A., Bessa, R., Keko, H., Carvalho, L., Issicaba, D., Sumaili, J., Miranda, V., 2011. Wind power forecasting uncertainty and unit commitment. *Applied Energy* 88, 4014–4023. <https://doi.org/10.1016/j.apenergy.2011.04.011>
- Wang, Y., Zhang, F., Kou, H., Zou, R., Hu, Q., Wang, J., 2025. A review of predictive uncertainty modeling techniques and evaluation metrics in probabilistic wind speed and wind power forecasting. *Applied Energy* 396, 126234. <https://doi.org/10.1016/j.apenergy.2025.126234>
- Wu, Z., Pan, S., Long, G., Jiang, J., Chang, X., Zhang, C., 2020. Connecting the dots: Multivariate time series forecasting with graph neural networks, in: Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 753–763. <https://doi.org/10.1145/3394486.3403118>
- Wu, Z., Pan, S., Long, G., Jiang, J., Zhang, C., 2019. Graph WaveNet for deep spatial-temporal graph modeling, in: Proceedings of the 28th International Joint Conference on Artificial Intelligence. pp. 1907–1913. <https://doi.org/10.24963/ijcai.2019/264>
- Yang, M., Huang, Y., Xu, C., 2025. Review of several key processes in wind power forecasting: Mathematical formulations, scientific problems, and logical relations. *Applied Energy* 377, 124631. <https://doi.org/10.1016/j.apenergy.2024.124631>
- Yu, B., Yin, H., Zhu, Z., 2018. Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting, in: Proceedings of the 27th International Joint Conference on Artificial Intelligence. pp. 3634–3640. <https://doi.org/10.24963/ijcai.2018/505>
- Yu, M., Zhang, Z., Li, X., Yu, J., Gao, J., Liu, Z., You, B., Zheng, X., Yu, R., 2020. Superposition

- graph neural network for offshore wind power prediction. *Future Generation Computer Systems* 113, 145–157. <https://doi.org/10.1016/j.future.2020.06.024>
- Zehtabiyani-Rezaie, N., Iosifidis, A., Abkar, M., 2023. Physics-guided machine learning for wind-farm power prediction: Toward interpretability and generalizability. *PRX Energy* 2, 013009. <https://doi.org/10.1103/PRXEnergy.2.013009>
- Zhang, Y., Wang, J., Wang, X., 2014. Review on probabilistic forecasting of wind power generation. *Renewable and Sustainable Energy Reviews* 32, 255–270. <https://doi.org/10.1016/j.rser.2014.01.033>
- Zhou, J., Lu, X., Xiao, Y., Tang, J., Su, J., Li, Y., Liu, J., Lyu, J., Ma, Y., Dou, D., 2024. SDWPF: A dataset for spatial dynamic wind power forecasting over a large turbine array. *Scientific Data* 11, 649. <https://doi.org/10.1038/s41597-024-03427-5>
- Zhou, Z., Botterud, A., Wang, J., Bessa, R.J., Keko, H., Sumaili, J., Miranda, V., 2013. Application of probabilistic wind power forecasting in electricity markets. *Wind Energy* 16, 321–338. <https://doi.org/10.1002/we.1496>

Appendix A. Training and implementation details

Training loop summary

The optimization procedure is identical across runs except for the routing penalties that are activated in each setting and the anchor labels that are available.

Algorithm A1 Physics-aligned MoE training

```

Input: history windows X, graph sequence A, targets Y, primary regime anchors R,
      auxiliary wake labels W (WTB only), model parameters theta
for each minibatch do
    Build node-level graph context from the precomputed graph tensors
    Encode X with the directed-diffusion GRU to obtain node context H
    Compute gate logits Z and soft routing probabilities G
    Compute expert forecasts and aggregate them with G
    Evaluate L_pred
    If MoE mode is active:
        add L_bal on top-K routing statistics
        add L_align on valid primary regime anchors
        if WTB: add L_force on MPPT/pitch samples
        if WTB: add L_aux on wake labels
        add L_smooth on the final graph snapshot
    Update theta with AdamW and mixed precision
end for
Select the checkpoint with the best validation RMSE

```

Notation and auxiliary losses

Table A1. Core notation used in the routing and loss definitions.

Symbol	Meaning	Dimension
$\mathbf{x}_{i,t}$	Node feature vector at node i , time t	$\mathbb{R}^F$
$\mathbf{X}_{t-H+1:t}$	Input history window	$\mathbb{R}^{H \times N \times F}$
$y_{i,t}$	Scalar target	$\mathbb{R}$
$\bar{p}_{i,t}$	Mean blade-pitch angle (Pab_mean) in WTB	scalar
$\text{Patv}_{i,t}$	Active power observed at anchor time; future values are prediction targets	scalar
$\mathbf{h}_{i,t}$	Encoded node context	$\mathbb{R}^d$
$\mathbf{a}_{i,t}$	Physics anchor vector used by the gate	$\mathbb{R}^q$
$\mathbf{z}_{i,t}$	Gate logits	$\mathbb{R}^E$
$\mathbf{g}_{i,t}$	Gate probabilities after softmax	$\mathbb{R}^E$
$f_e(\cdot)$	Expert- e prediction head	$\mathbb{R}^d \rightarrow \mathbb{R}^P$
$R_{i,t}$	Primary regime anchor label	discrete
$W_{i,t}$	Auxiliary wake flag in WTB	$\{0, 1\}$
$\mathcal{A}_t(i, j)$	Edge weight from node j to node i at time t	scalar
E	Number of experts	scalar
K	Top- K used in the balancing loss	scalar

Let $\mathcal{B}$ denote the routed node-anchor samples in a mini-batch after target and anchor masks are applied, and let $B = |\mathcal{B}|$. For sample n , the soft importance and normalized share of expert e are

$$I_e = \frac{1}{B} \sum_{n=1}^B g_n^{(e)}, \quad P_e = \frac{I_e}{\sum_{r=1}^E I_r}.$$

The empirical top- K load is

$$f_e = \frac{1}{B} \sum_{n=1}^B \mathbf{1}[e \in \text{TopK}(\mathbf{g}_n)],$$

and the balancing loss is

$$\mathcal{L}_{\text{bal}} = E \sum_{e=1}^E f_e P_e - 1.$$

For WTB wake supervision,

$$\mathcal{L}_{\text{aux}} = \frac{1}{|\Omega_{\text{wake}}|} \sum_{(i,t) \in \Omega_{\text{wake}}} \text{BCE}(z_{i,t}^{(\text{wake})}, W_{i,t}),$$

where Ω_{wake} contains only MPPT and pitch-control samples for which the wake flag is defined. The graph-smoothness penalty is

$$\mathcal{L}_{\text{smooth}} = \frac{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) \|\mathbf{g}_{i,t} - \mathbf{g}_{j,t}\|_2^2}{\sum_{t \in \mathcal{T}_{\mathcal{B}}} \sum_{i,j} \mathcal{A}_t(i,j) + \epsilon}.$$

Graph construction details

For WTB, each turbine has a fixed coordinate $\mathbf{p}_i = (x_i, y_i)$. A candidate set is first built by retaining the k_{nn} nearest turbines within a maximum radius d_{max} . Let $\Delta \mathbf{p}_{ij} = \mathbf{p}_j - \mathbf{p}_i$ and let $\theta_{i,t}$ denote the local meteorological wind-from direction. The downstream unit vector is

$$\mathbf{u}_{i,t} = \begin{bmatrix} \sin(\theta_{i,t} + \pi) \\ \cos(\theta_{i,t} + \pi) \end{bmatrix}.$$

This gives

$$d_{ij,t}^{\parallel} = \Delta \mathbf{p}_{ij}^{\top} \mathbf{u}_{i,t}, \quad d_{ij,t}^{\perp} = |\Delta p_{ij}^x u_{i,t}^y - \Delta p_{ij}^y u_{i,t}^x|.$$

An edge is activated only when turbine j falls inside a downstream cone:

$$\mathbb{I}_{ij,t}^{\text{cone}} = \mathbf{1} \left[d_{ij,t}^{\parallel} > 0 \wedge \arctan \left(\frac{d_{ij,t}^{\perp}}{\max(d_{ij,t}^{\parallel}, 10^{-6})} \right) \leq \phi \right].$$

The wake weight is

$$\tilde{\mathcal{A}}_t(i,j) = \exp \left(-\frac{d_{ij,t}^{\parallel}}{\alpha} \right) \exp \left(-\frac{|d_{ij,t}^{\perp}|}{\beta} \right) \mathbb{I}_{ij,t}^{\text{cone}}.$$

If wind direction is missing, the implementation falls back to a static proximity weight,

$$\mathcal{A}^{\text{static}}(i,j) = \exp \left(-\frac{\|\Delta \mathbf{p}_{ij}\|_2}{d_{\text{max}}} \right).$$

Only the strongest M inbound weights are retained for each target and time step. The wake score is

$$s_{i,t}^{\text{wake}} = \sum_j \mathcal{A}_t(i, j),$$

and the auxiliary wake flag is

$$W_{i,t} = \begin{cases} 1, & \text{if } s_{i,t}^{\text{wake}} \geq q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ 0, & \text{if } s_{i,t}^{\text{wake}} < q_{0.75}^{\text{wake}} \wedge R_{i,t}^{\text{wtb}} \in \{1, 2\}, \\ \emptyset, & \text{otherwise.} \end{cases}$$

ERA5 treats each grid point as a node with geographic coordinate (φ_i, λ_i) and computes great-circle distance with the Haversine formula

$$d_{ij} = 2R \arctan\left(\frac{\sqrt{a_{ij}}}{\sqrt{1 - a_{ij}}}\right),$$

where $R = 6371$ km and

$$a_{ij} = \sin^2\left(\frac{\varphi_j - \varphi_i}{2}\right) + \cos(\varphi_i) \cos(\varphi_j) \sin^2\left(\frac{\lambda_j - \lambda_i}{2}\right).$$

The retained symmetric graph is

$$\mathcal{A}(i, j) = \exp\left(-\frac{d_{ij}^2}{2\sigma^2}\right) \mathbf{1}[j \in \mathcal{N}_{k_{\text{nn}}}(i) \text{ or } i \in \mathcal{N}_{k_{\text{nn}}}(j)],$$

where σ is the median retained neighbor distance on the training graph.

Shared constants

Table A2. Shared architecture, graph, and training constants used in the reported experiments.

Item	WTB	ERA5	Role
History / horizon (H, P)	(36, 24)	(36, 24)	Input and forecast lengths
Gate softmax temperature τ	0.7	0.7	Soft routing sharpness
Top- K in L_{bal}	1	1	Balancing statistic only
Hidden size d	64	64	Shared encoder width
k_{nn} candidates	8	8	Initial neighborhood size
Maximum candidate radius d_{max}	1500 m	not used	WTB proximity fallback / candidate filter
Wake-cone half-angle ϕ	25°	not used	WTB downstream activation
Streamwise decay α	1200 m	not used	WTB wake attenuation
Cross-stream decay β	400 m	not used	WTB wake attenuation
Retained inbound edges M	5	symmetric k_{nn} graph	Final graph sparsity
Optimizer	AdamW	AdamW	Shared optimizer
Learning rate	2×10^{-3}	2×10^{-3}	Shared schedule
Weight decay	10^{-4}	10^{-4}	Shared regularization
Batch size	16	16	Shared mini-batch size
Gradient clipping	1.0	1.0	Shared training stabilization

Dataset-specific thresholds and routing weights

Table A3. Dataset-specific regime thresholds used to construct routing anchors.

Dataset	Threshold	Value and meaning
WTB	u_{idle}	3.0 m s ⁻¹ , idle / active split
WTB	u_{rated}	10.5 m s ⁻¹ , MPPT / pitch boundary
WTB	p_{th}	2.0 deg, pitch-angle threshold
WTB	$q_{0.75}^{\text{wake}}$	1.0690, upper-quartile wake-score threshold
ERA5	$\varepsilon_{\text{sshf}}$	169,116, 10th percentile of $ \text{sshf} $ on the training split
ERA5	$q_{0.95}^{\Delta}$	458,287.125, 95th percentile of $ \Delta \text{sshf} $ on the training split

Table A4. Active routing-loss weights in the reported corrected models.

Weight	WTB	ERA5	Role
λ_{bal}	1000	0.01	Expert-usage stabilization
λ_{align}	5000	0.2	Coarse regime alignment
λ_{force}	10000	0	Boundary-focused forcing
λ_{aux}	250	0	Wake-sensitive auxiliary routing
λ_{smooth}	0.05	0.05	Local routing coherence